A Machine-Checked Audit of the Extended-Extraction RG Contraction in the Wilson-Lattice / OS Route to a 4D Yang–Mills Mass Gap

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Abstract

Ben Goertzel's manuscript *Yang–Mills Mass Gap in 4D: RG Bootstrap, Polymer Expansion, OS Reconstruction* (v14) proposes to construct a continuum SU(N) Yang–Mills theory from Wilson lattice gauge theory via a multiscale renormalization-group (RG) bootstrap, a Kotecký–Preiss (KP) polymer expansion, exponential clustering, and Osterwalder–Schrader (OS) reconstruction. Its technical keystone is an *extended-extraction* contraction $\Lambda = C_1\lambda_{\mathrm{irr}} < 1$, with advertised constant near 2224 and $\lambda_{\mathrm{irr}} = 2^{-13}$ at block factor 2 and extraction depth 16.

The scalar bootstrap arithmetic is exact, but the advertised Wilson recursion constant is *not derived in v14 as written.* The main-text factor product is 33152 at $b = 2$; Appendix O's different proposed factorization is 11088/5 when its extraction factor is 2. A sign-only repair of the printed extraction series is insufficient once the displayed multivariate multiplicities are retained.

We construct a concrete finite Wilson-polynomial realization on actual group-valued link fields. In its explicit coefficient-majorant norm, canonical-dimension truncation is gauge invariant and has operator constant 1. This does not transfer from v14's range-only specification of P_{ext}: a skew projection onto the same low-coordinate range has arbitrarily large coefficient norm. We prove the correct transfer theorem from explicit basis and dual-jet matrices with their column-sum conditioning bounds, but v14 does not provide those data for the dimension-16 action sector. A direct source reconciliation shows that they cannot be reconstructed uniquely from the present text: Appendix F.4 uses equations of motion to identify the operator space, while Appendix O.9 explicitly retains EOM operators in an off-shell basis; neither section supplies lattice Wilson representatives or a quotient basis, and Appendix I's explicit dual jets stop at the distinct dimension-4 observable sector. Likewise, regrouping connected families by union support has norm 1; the factor 14 is exactly a *local rooted branching count* $2(2 \cdot 4 - 1)$, not a bound on the total contacts of two extended polymers. The fluctuation big-O and weighted cumulant-generation estimates also require quantitative proofs on the actual Wilson RG map.

A third machine-checked round works on the exact SU(2), $b = 2$ Wilson block with normalized Haar measure and decides the extraction stage from three sides. (i) Representative-wise hard truncation is not well-defined on the actual trace-word quotient: for the centered trace words $a = 2 - \mathrm{Tr}(U)$ and $b = 2 - \mathrm{Tr}(U^2)$, Cayley–Hamilton forces $a^3b - 4a^4 + a^5 = 0$ with monomial canonical dimensions 16, 16, 20; truncation at 16 deletes only a^5, and the truncated expression evaluates to -32 on the holonomy $\mathrm{diag}(i, -i)$ while the exact expression vanishes identically. (ii) Two corrected constructions nevertheless carry exact norm-below-two extractors — the full associated-graded jet chart ($131071/65536 < 2$) and the one-plaquette trace quotient ($31/16 < 2$); with Appendix O's other factors kept unchanged, the resulting conditional constant $90832203/40960 \approx 2217.6$ closes the two-source bootstrap at depth 16 and fails at depth 15. (iii) Neither survivor transfers to v14's canonical (F, D) operator space: a dimension-18 scalar descendant has radial degree only 6 (resp. 3), so both extractors retain it, and the dimension-6

descendant $\partial^2\mathrm{Tr}(F^2)$ is zero in the manuscript's integration-by-parts quotient yet has a nonzero gauge-invariant realization on an explicit three-plaquette block. The transfer socket required before any Wilson-RG intertwiner is therefore uninhabited as written, under both the on-shell and off-shell relation policies.

A fourth machine-checked round tests the replacement census against the exact symmetry of the finite-spacing lattice. On an explicit SU(2) Cartan link chart, the sum of squared normalized Wilson potentials over the six coordinate plaquettes is gauge invariant and has canonical dimension 8. Its exact fourth jet is $6\sum_{\mu<\nu} F_{\mu\nu}^4$. A rational orthogonal rotation changes the corresponding quartic from 1 to 337/625. Thus the observable is hypercubic-invariant but not $O(4)$-invariant. Every span of Appendix O.9.3's Lorentz-scalar contractions remains $O(4)$-invariant, so it cannot be complete for actual lattice Wilson observables. This obstruction closes the off-shell, on-shell EOM, Bianchi/IBP, trace, Cayley–Hamilton, and postponed-IBP variants of the same Lorentz-only target. It does not rule out a finite coordinate system: it forces either a genuinely hypercubic census at finite lattice spacing or a separately proved continuum/Symanzik projection with a quantitatively controlled Lorentz-breaking remainder.

A fifth machine-checked round begins the hypercubic branch. On the unpadded derivative-free Cartan quartic carrier, all 126 monomials are classified by signed $H(4)$ orbit certificates. Exactly four admissible orbit basis functions remain; evaluation at their representatives gives four duals and the identity conditioning matrix, and every hypercubic-covariant rational quartic coefficient function is reconstructed from them. The exact Wilson fourth jet has coordinate vector $(6, 0, 0, 0)$ in this basis. The subspace of curvature observables invariant under every $O(4)$ transformation is proper, and the pure-fourth Wilson polynomial is a nonzero class in the corresponding Lorentz-breaking quotient. The attempted dimension-16 extension then reaches a different terminal obstruction: its fixed-width raw carrier retains inactive padding as part of equality. Two distinct encodings with zero active fields and derivatives have identical active syntax. Any basis faithful to active syntax must give them equal rows, whereas the proposed exact census requires a right inverse and hence distinct identity-matrix rows. Lean proves that no invertible exact census on this padded carrier can have every basis column padding-invariant. The next finite-spacing construction must use an unpadded dependent syntax or an explicit padding quotient before relation rows and dimension-16 duals are meaningful.

Ours: the unpadded dimension-16 replacement is now constructed and connected by an explicit equivalence to the exact-dimension subtype of the active (F, D) syntax. Its raw carrier has 23,027,949,358,636,282,675,200 elements. Character averaging over all 384 signed coordinate permutations gives rank-16 invariant tensor multiplicities 22,379,179 (scalar) and 22,368,256 (pseudoscalar); after the exact 5,361,612,271,200 trace/derivative layouts are included, the raw invariant coefficient spaces have dimensions 119,988,480,745,781,344,800 and 119,929,915,854,943,027,200. These are exact raw-carrier dimensions before antisymmetry, Bianchi, EOM, integration by parts, trace, or Cayley–Hamilton rows, not a physical operator basis. After the exact derivative-localization, field-strength antisymmetry, and field-label quotients, ordinary permutation conjugacy classes give a second exact character calculation: the resulting finite local-word orbit carrier has scalar and pseudoscalar $H(4)$ invariant multiplicities 246,136,792 and 246,121,092. The proof identifies every permutation-partition fiber with its ordinary conjugacy class and cancels its class cardinality against the centralizer factor, so no representative or padding convention enters these values. They still precede Bianchi, EOM, IBP, commutator, tracelessness, and matrix trace relations and are not a physical operator basis. On the separately complete derivative-free Cartan octic sector, 1287 monomials reduce to 17 signed-orbit coordinates with exact duals and identity conditioning. The actual Wilson eighth jet occupies only the pure-octic coordinate, with Symanzik coefficient $-a^{12}/20160$. For bounded constant Cartan curvature, the normalized Wilson density differs from its $O(4)$-invariant quadratic term by at most $(5/8)a^4M^4$ and converges to that term as $a \to 0$. The two-block boundary-cochain lift retains extraction constant 1; carrying the three still-conditional analytic factors gives budget 5544/5, whose two-source arithmetic closes first at depth 15. This is a quantitative tree-level constant-Cartan restoration theorem, not an interacting Symanzik theorem or a constructive continuum limit.

A sixth machine-checked round makes the repaired finite relation target coordinate-free. For either relation policy, the invariant physical relation submodule $\mathcal{J}_p$ is defined as the preimage of the full physical relation range inside the signed $H(4)$-invariant orbit space. The exact off-shell-to-on-shell inclusion is proved. A family of 11,556 transferred field-eight trace relations, an independent seven-to-eight commutator row, and an explicit four-to-five-field commutator row with Reynolds coordinate $-1/24$ give the certified bound

$$11558 \leq \dim_{\mathbb{Q}} \mathcal{J}_p.$$

This is a lower bound, not a complete joint rank, quotient dimension, or conditioning certificate. Our finite derivative-alpha-reduced F, D/IBP relation cochains now map by the physical relation operator followed by signed $H(4)$ Reynolds averaging onto this same J_p. The map is $H(4)$-equivariant, surjective for both policies, and has range equal to the whole J_p; shared off-shell cochains retain their ambient invariant semantics on shell. This identifies the repaired finite relation object. It does not supply the still-missing full noncommutative Wilson-functional coordinates or analytic dual jets. A further exact comparison fixes why the existing Cartan chart cannot be silently reused for that task: the two literal eight-field endpoints of a certified incoming covariant-commutator column have the same unordered Cartan plane decoration but different trace orders. Our raw Cartan restriction of the actual Wilson eighth-jet coefficient kills their signed difference, while its ordinary derivative-free trace-quotient class is certified nonzero. Independently, that trace quotient has dimension 98, so no map from it into a 17-coordinate Cartan chart can be injective. More sharply, no ordinary-trace-quotient coordinate which agrees with the actual Cartan eighth-jet scalar on every raw labelled input can even detect that certified incoming trace-order class; an injective realization of this scalar coordinate is therefore formally uninhabited. This is not a no-go for the Wilson route. It specifies the repair: a full Wilson coordinate system must retain noncommutative trace order and the coupled lower-field commutator component in additional coordinates (`HypercubicDimension16WilsonCartanJointMismatch`, `HypercubicDimension16WilsonCartanTraceOrderFaithfulnessNoGo`).

Ours. The first finite repair requirement is now exact. On the concrete seven/eight-field commutator quotient, the canonical eight-field trace coordinate cannot descend by itself. More sharply, every descended trace-order correction has a forced nonzero value on the seven-field component of the coupled row. The explicit compensator realizes that value; it kills the complete projected physical commutator for either relation policy while still detecting its isolated eight-field trace component. We package this rational quotient coordinate beside the exact Cartan Wilson coefficient packet, not as an identification of the two and not as a completed analytic noncommutative Wilson functional (`HypercubicDimension16WilsonTraceOrderRepair`).

Ours. The trace-order layer is now fully coordinatized on this bounded quotient. The existing 98 exact ordinary trace duals assemble into an injective packet with a checked synthesis map recovering every ordinary trace class. Composed with the forced seven-field compensator, this packet kills the full projected physical commutator for either relation policy while retaining its isolated eight-field component as the exact nonzero two-pivot trace vector. This makes a finite noncommutative trace-order coordinate layer available beside the Cartan packet; it does not identify that rational layer with an analytic Wilson functional, establish descent through the full relation range, or supply a conditioning estimate for the added seven-field compensator (`HypercubicDimension16WilsonTraceOrderCoordinateCompletion`).

Ours. The first signed-hypercubic extension is then tested rather than assumed. A one-coordinate extension kills the commutator and derivative-free trace rows but has an explicit nonzero value on a physical IBP row. A second field-relabel-invariant split-derivative/plane coordinate supplies exactly the missing independent value. The resulting two-coordinate signed-$H(4)$-invariant correction kills the concrete commutator, the displayed IBP row, and every lifted derivative-free eight-field trace generator. A further exact test constructs a genuine three-distinct-axis innermost Bianchi row; both coordinates have zero signed weights on its three normalized summands, so the correction kills the row under either policy and its Reynolds-averaged invariant relation member. An explicit on-shell EOM row has four contracted-axis terms; both coordinates have zero signed weights on each term, while exact normalization removes the diagonal term and leaves orientations $1, 1, -1$. Thus the correction also kills that physical row and its Reynolds-averaged on-shell relation member.

Ours. The next genuine IBP row is not silently absorbed. It has a different split derivative pattern, and the two-coordinate correction has the exact nonzero value (1/2) times the incoming trace class on it. A third field-relabel-invariant plane/derivative coordinate has calibrated value 1 on this row and signed weight 0 on the already checked commutator, first IBP, Bianchi, and EOM rows. Subtracting one half of its trace contribution therefore kills the new physical row and its Reynolds-averaged invariant relation member while retaining those earlier physical and invariant checks. This is a finite repair family: it neither establishes annihilation of the full Bianchi/EOM/IBP/trace relation range nor constructs noncommutative Wilson-functional analytic coordinates (`HypercubicDimension16WilsonTraceOrderInvariantRepair`, `HypercubicDimension16WilsonTraceOrderInvariantBianchi`, `HypercubicDimension16WilsonTraceOrderInvariantEOM`, `HypercubicDimension16WilsonTraceOrderInvariantThreeCoordinateRepair`).

Ours. The next canonical trace family supplies an exact boundary test rather than an implicit closure claim. On the same decorated seven-field carrier, the polarized SU(2) trace-anticommutator row has source, adjacent-swap, and split-trace terms. The three existing coordinates see the same field and derivative decorations in all three terms: the first has total value 1/2, while the two plane/derivative coordinates vanish. The three-coordinate correction therefore has the exact nonzero value (1/2) times the incoming trace class, both on the physical row and on its Reynolds-averaged member of J_p, for either policy. Thus it cannot descend through the complete relation range (`HypercubicDimension16WilsonTraceOrderTraceAnticommutatorObstruction`).

Ours. That failure has a finite, trace-topological repair. We gate the derivative-axis character by the relabel-invariant condition that the trace permutation has exactly one two-cycle. The source and swap have cycle type $\{7\}$, while the split term has type $\{2, 5\}$; their exact signed weights are respectively 0, 0, and 32. The resulting fourth scalar is -1 on the full trace-anticommutator row, so adding one half of its incoming trace-class contribution cancels the preceding (1/2) residual. Kernel replay also gives zero on the displayed commutator, both explicit IBP rows, Bianchi, on-shell EOM, and every lifted derivative-free eight-field trace row, before and after Reynolds averaging. This repairs the displayed finite family only: it does not establish descent through all of J_p, a complete relation census, or a noncommutative Wilson-functional coordinate system (`HypercubicDimension16WilsonTraceOrderTraceTopologyRepair`).

Ours. The canonical fundamental three-cut identity tests that repair on a distinct six-term trace wiring. Its six summands have cycle types $\{7\}, \{6\}, \{2, 5\}, \{6\}, \{7\}, \{5\}$. The four-coordinate functional therefore has exact value $-\frac{1}{2}$ times the incoming trace class, on both the physical row and its Reynolds member. A fifth scalar gates the same derivative-axis character by the presence of one six-cycle. Its exact signed weights on those six summands are $0, 32, 0, 32, 0, 0$, giving normalized value -2; adding $-\frac{1}{4}$ of its incoming trace contribution cancels the residual. The new scalar has zero value on the displayed trace-anticommutator row, so the two named canonical families remain killed before and after averaging. This is again a finite named-family repair, not a proof of descent through all of J_p or a complete Wilson-functional coordinate construction (`HypercubicDimension16WilsonTraceOrderFundamentalThreeCutRepair`).

Ours. The two individual gate repairs admit a more constrained finite continuation. We enumerate all 210 ordered choices of three distinct cut labels on the same seven-cycle, form the alternating six-term trace-cycle profile, and find four topology equations. Together with the polarized trace-anticommutator equation, they uniquely force the coefficients

$$(c_2, c_3, c_4, c_5, c_6) = \left(\tfrac{6}{5}, \tfrac{13}{10}, -1, -\tfrac{7}{10}, -\tfrac{3}{5}\right)$$

for the gates recording one cycle of lengths two through six. Kernel replay checks that this full rational profile vanishes for every one of the 210 ordered placements. We lift it to a signed-hypercubic coordinate and verify that it kills the two displayed canonical physical rows and their Reynolds-averaged members. Each certified placement is now mapped to a distinct actual fundamental-trace site and policy-indexed physical relation generator; its finitely supported source cochains embed injectively in our F, D/IBP cochain space and map by signed Reynolds semantics to the coordinate-free joint relation submodule. The canonical label recovers the previously displayed fundamental row. A separate symbolic field-sector transport proves that the whole finitely supported physical three-cut family has zero exact field-eight invariant projection for either relation policy: all six trace-rewire terms remain seven-field. OUR physical trace-cycle bridge next factors every rewire's signed Reynolds weight through the same profile gate, replays the six-gate cancellation on all 210 actual labels, and proves that the invariant trace-cycle-profile coordinate annihilates every finitely supported cochain in this physical family under either policy. The three earlier coordinate weights are trace-order blind: rewiring changes the trace but not the decorated local word, so the six signed weights agree and their alternating coefficients cancel identically. We transport that identity through the actual physical relation operator and prove that the separate three-coordinate correction also annihilates every cochain in this family. Thus the full current cycle-profile correction is zero on this realized physical family under either policy. This closes the field-eight sector and all current correction coordinates for this family only; it does not identify every physical source generator, establish descent through all of J_p, give a complete relation census or conditioning certificate, or construct noncommutative Wilson-functional coordinates (`HypercubicDimension16FundamentalTraceCycleProfile`, `HypercubicDimension16WilsonTraceOrderCycleProfileRepair`, `HypercubicDimension16FundamentalTracePhysicalFamily`, `HypercubicDimension16FundamentalTracePhysicalFieldEight`, `HypercubicDimension16FundamentalTracePhysicalCycleProfile`, `HypercubicDimension16FundamentalTracePhysicalThreeCoordinate`).

Ours. The polarized trace-anticommutator test is now extended from its displayed placement to all seven adjacent placements on the same actual physical seven-cycle. Each physical row has the unrewired, adjacent-swap, and split-trace terms with one common field/derivative decoration. The three trace-order-blind coordinates consequently retain the common $\frac{1}{2}$ incoming trace-class residual. A kernel-reduced finite replay of the seven trace topologies gives the common profile value $-\frac{1}{2}$, so the profile contribution cancels that residual at every placement. The seven basis rows embed injectively into the policy-indexed F, D/IBP source and their finitely supported cochains map by signed Reynolds semantics into the coordinate-free joint relation submodule; the full current cycle-profile correction vanishes on this whole finite family under both policies. This is a second realized trace-family block, not a census of trace-anticommutator sites on arbitrary carriers or a descent theorem for all of J_p (`HypercubicDimension16TraceAnticommutatorPhysicalFamily`).

Ours. The full profile correction is also tested against the five displayed non-trace calibration rows of the physical relation operator: the covariant commutator, the original and second integration-by-parts rows, Bianchi, and the on-shell equation-of-motion row. For every normalized seven-field summand, the signed $H(4)$ orbit sum of the new profile multiplicity is replayed exactly in the kernel; each relevant sum is zero. Linearity and the already-identified physical rows then show that the full correction kills both the physical rows and their Reynolds-averaged invariant representatives. Those representatives are genuine members of the coordinate-free joint relation submodule. An arbitrary-carrier bridge then identifies every physical source-eight traceless, polarized-anticommutator, and fundamental three-cut row with the derivative-free field-eight schema. The derivative-bearing site types are empty at that source count. Thus the full correction annihilates every source-eight physical relation column, not only a chosen lifted family. OUR source-six curvature-pair bridge next handles the sole cross-sector covariant-commutator branch: its two seven-field insertions have the same three decoration coordinates and conjugate trace permutations, hence the same signed-Reynolds profile coordinate, so their opposite row signs cancel. Combined with the field-support theorem below six, the full correction now annihilates every arbitrary physical relation column outside the source-seven band. This is not a source-seven census, a complete physical relation census, a proof that these rows span J_p, or descent through all of J_p (`HypercubicDimension16WilsonTraceOrderCycleProfileSourceSix`, `HypercubicDimension16FieldEightTracePhysicalCompleteness`).

Ours. The remaining source-seven boundary is now narrowed by an exact counterexample to the present *one-class* repair, rather than left as an undifferentiated census task. We construct a second physical covariant-commutator carrier whose two curvature insertions have plane profile $(0, 0, 1, 0, 0, 0, 0, 1)$ and different trace orders. Its signed eight-field pair has a kernel-checked value $-4/3$ under an ordinary derivative-free trace quotient functional. At the same evaluation assignment, the original incoming class used by each of the existing seven-field compensator coordinates has value zero. Consequently the functional evaluates the full current cycle-profile correction on this actual row to $4/3$, so that correction is nonzero under both physical relation policies. This refutes only the current correction family, whose compensators all take values in the single original incoming trace class. We then construct a bounded two-class repair of these two actual rows. Two field-relabel-invariant signed-$H(4)$ selectors count a whole two-derivative word together with the plane on a distinct field. Their exact values on (incoming, secondary) are $(1, 0)$ and $(0, 1)$, respectively. Rather than substitute that standalone map for the prior correction, we replay the new secondary selector on the already checked first-IBP, Bianchi, on-shell EOM, fundamental three-cut, polarized trace, and lifted field-eight rows; every listed value is zero. The genuine second-IBP row instead has normalized selector value 1. Let R be the exact (and certified nonzero) residual of the prior correction on the secondary row. OUR two-class augmentation is the prior correction minus the secondary selector times R; on the second-IBP row it has exact value $-R$, so this is a machine-checked incompatibility rather than an omitted check. We then use the independent third split-derivative/plane coordinate already calibrated by the earlier three-coordinate construction. Its signed sums vanish on both displayed source-seven commutators and its normalized second-IBP value is 1. Adding this third coordinate times R produces OUR three-class correction. It cancels both commutators and the named first-IBP, second-IBP, Bianchi, on-shell EOM, fundamental, polarized-trace, and lifted field-eight rows under either relation policy. A further actual source-seven singleton-trace row then refutes this three-class architecture under both policies. It shares the incoming two-derivative word but has one genuine singleton trace. Its exact signed weights for the original derivative, IBP, third, and cycle-profile coordinates are 32, 0, 0, and $-96/5$, respectively; both plane-profile selectors vanish. Hence the full three-class correction has exact value $(-1/10)$ times the certified nonzero incoming trace class. More generally, every correction obtained from the cycle-profile term by assigning arbitrary quotient values to these same three selectors has that value on this row, so no adjustment of those three added selector values, with the cycle-profile base held fixed, can descend through the physical relation submodule. The next finite task is to solve the exact compatibility system after also allowing the existing base scalar coordinates to vary. This is a machine-checked mismatch, not a complete source-seven census, joint rank or conditioning calculation, or Wilson-functional coordinate system (`HypercubicDimension16WilsonTraceOrderCycleProfileSourceSevenMismatch`, `HypercubicDimension16WilsonTraceOrderCycleProfileSourceSevenCorrectionRefutation`, `HypercubicDimension16WilsonTraceOrderCycleProfileSourceSevenTwoClassRepair`, `HypercubicDimension16WilsonTraceOrderCycleProfileSourceSevenTwoClassCompatibility`, `HypercubicDimension16WilsonTraceOrderCycleProfileSourceSevenThreeClassRepair`, `HypercubicDimension16WilsonTraceOrderCycleProfileSourceSevenThreeClassTracelessRefutation`).

Ours: the boundary-cochain, hypercubic, and joint-quotient constructions repair precise broken steps within the same Wilson lattice $\to$ extended-extraction RG $\to$ KP $\to$ glue $\to$ OS proof programme. They are not an alternative endpoint theorem, and they are not attributed to v14 where v14 does not contain them.

Thus there is presently no derived constant C_{true} for the Wilson recursion. The support recurrence, two-marked glue identity, stopping-scale polymer estimates, Wilson reflection positivity, and continuum/OS construction remain route-specific open inputs. The finite $\mathbb{Z}_2$, $\mathbb{Z}_3$, and Q_8 gaps checked in Lean are real but Clay-irrelevant (single-link, strong-coupling, finite). The Yang–Mills mass gap remains open; we make no claim about it.

1 The problem and Ben Goertzel's route

1.1 The Millennium problem and the lattice strategy

Fix the gauge group $G = \mathrm{SU}(N)$ with $N \geq 2$ and let $\mathbb{R}^4$ be Euclidean space. The Yang–Mills mass-gap problem asks for the construction of a nontrivial continuum quantum Yang–Mills theory on $\mathbb{R}^4$ with gauge group G, satisfying the Wightman / Osterwalder–Schrader axioms, whose reconstructed Hamiltonian has a unique vacuum and a strictly positive spectral gap on the orthocomplement of the vacuum.

Goertzel's v14 manuscript follows Wilson's lattice regularization. One discretizes $\mathbb{R}^4$ as a hypercubic lattice $a\mathbb{Z}^4$ with spacing $a > 0$, with link variables $U_{x,\mu} \in G$ on oriented nearest-neighbor links. The plaquette variable is the parallel transporter around an elementary square,

$$U_{x,\mu\nu} \;=\; U_{x,\mu}\, U_{x+ae_\mu,\nu}\, U^{-1}_{x+ae_\nu,\mu}\, U^{-1}_{x,\nu}, \qquad 1 \leq \mu < \nu \leq 4,$$

and the Wilson action penalizes deviation of plaquettes from the identity,

$$S_{\Lambda,a}(U) \;=\; \beta \sum_{p \subset \Lambda} V(U_p), \qquad \beta = \frac{2N}{g^2}, \qquad V(U) := 1 - \tfrac{1}{N}\mathrm{Re}\,\mathrm{Tr}(U) \in [0,2].$$

Weak coupling is $\beta \gg 1$. The strategy is: (Step 1) start from the well-defined finite-dimensional lattice integral; (Step 2) run a multiscale RG bootstrap controlling the effective theory across all scales; (Step 3) establish exponential clustering at a fixed physical scale ℓ_* via a KP polymer expansion; (Step 4) transfer clustering from the coarse scale back to the fine scale with a glue / back-propagation lemma; (Step 5) reconstruct the continuum Hilbert space, vacuum, and Hamiltonian by OS, with the clustering rate becoming the spectral gap; (Step 6) verify nontriviality through a nonzero third cumulant of the action density.

1.2 The continuum-limit obstruction and extended extraction

In any RG scheme one must show that the "irrelevant" sector (operators of canonical dimension > 4) contracts at each step:

$$\|K^{\mathrm{irr}}_{j+1}\| \;\leq\; \Lambda\, \|K^{\mathrm{irr}}_j\| + (\text{source terms}), \qquad \Lambda < 1.$$

The manuscript identifies a sharp obstruction in the *naive* scheme, where one extracts only the relevant and marginal operators of dimension ≤ 4. Under rescaling, a dimension-d operator scales by b^{4-d}, so the leading irrelevant operators (dimension 6) contract by $b^{-2} = 0.25$ for $b = 2$. But the RG step also recombines polymers, contributing a combinatorial constant C_1. The manuscript advertises the working bound $C_1 \leq 2224$ for $b = 2$; if that value were established, the naive gain would be

$$\Lambda_{\mathrm{naive}} \;=\; 2224\, b^{-2} \;=\; 2224 \times 0.25 \;=\; 556 \;>\; 1,$$

an *expanding* map: initial smallness would grow like 556^J over $J \sim |\log a_0|$ steps and diverge as $a_0 \to 0$.

The proposed resolution is *extended extraction.* Instead of extracting only dimension ≤ 4 operators, one extracts all gauge-invariant local operators up to canonical dimension $d_{\max} = 16$ (for $b = 2$). The irrelevant sector then begins at dimension $d_{\max} + 1 = 17$ and contracts by

$$\lambda_{\mathrm{irr}} \;=\; b^{4-(d_{\max}+1)} \;=\; b^{3-d_{\max}} \;=\; 2^{-13} \;\approx\; 1.2 \times 10^{-4}.$$

With the same advertised value, the proposed contraction factor becomes

$$\boxed{\Lambda \;=\; C_1\,\lambda_{\rm irr} \;\le\; 2224\cdot 2^{-13} \;\approx\; 0.27 \;<\; 1.}$$

The manuscript presents this as genuine contraction "with substantial slack." The audit below verifies this arithmetic only *conditional on the supplied constant*; it refutes the available derivations of that constant as written. The price is tracking a finite-dimensional vector of higher couplings $\vec{g}_j = (g_j^{(4)}, g_j^{(6)}, \ldots, g_j^{(d_{\max})})$ rather than the single gauge coupling; the manuscript argues these are perturbatively small and do not affect the universal one-loop beta-function coefficient $b_0 = 11/(3\cdot 16\pi^2)$, which is determined entirely by local ultraviolet information.

1.3 The recombination constant and the extraction-projection norm

Appendix O proposes the factorization

$$C_1 \;\le\; C_{\rm fluct}\cdot C_{\rm cumulant}\cdot C_{\rm recomb}\cdot C_{\rm extract} \;\le\; 1.65\times 3b^4\times 14\times 2,$$

which at $b = 2$ equals $11088/5 = 2217.6$, rather than exactly 2224. Appendix F.5 displays a different factor product which equals 33152 at $b = 2$; the two ledgers are inconsistent. The last Appendix-O factor $C_{\rm extract} = \|P_{\le d_{\max}}\|_{\rm op}$ is the operator norm of the extraction projection onto the finite-dimensional space of operators of dimension $\le d_{\max}$. The manuscript bounds it (its Computation O.7(d)) by

$$\|P_{\le d_{\max}}\|_{\rm op} \;\le\; \sum_{d=0}^{d_{\max}} \binom{\#\text{indices}}{d} (r_1/r_0)^{-d} \;\le\; 1 + O\big((r_0/r_1)^{d_{\max}}\big),$$

and asserts $\|P_{\le d_{\max}}\|_{\rm op} \le 2$ for $r_1/r_0 = 1/2$ and $d_{\max} = 16$. The literal negative exponent grows, but changing only its sign does not turn the displayed multivariate sum into a one-dimensional geometric series. More fundamentally, v14 specifies $P_{\rm ext}$ by its range and does not give the dimension-16 invariant basis, dual jets, or conditioning estimates needed to determine its operator norm. We return to the proved realization and the transfer obstruction in §3.

2 What we formalized

The formalization lives in the Lean namespace `Mettapedia.QuantumTheory.YangMills`. We summarize the four checked components: the contraction predicate, the knife-edge arithmetic, the finite strong-coupling gaps with their finite OS reconstruction, and the continuum-open gating. All theorem and definition names below are reproduced verbatim from the corresponding modules.

2.1 The extended-extraction contraction predicate

The module `Mettapedia.QuantumTheory.YangMills.RGBootstrap` models the route's contraction obligation as a predicate over a recombination constant C, block factor b, and extraction depth $d_{\max}$. The irrelevant scaling factor is $b^{3-d_{\max}}$ (an integer power, since the useful cases have $d_{\max} > 3$), and a contraction is a nonnegative constant whose product with the irrelevant scale is below 1.

```
/-- Irrelevant scaling factor ‘b^(3-dmax)‘ used by the extended-extraction
discussion.  The exponent is an integer because the useful cases have
‘dmax > 3‘. -/
def irrelevantScale (b dmax : ℕ) : ℝ :=
```

```
  (b : ℝ) ^ (3 - (dmax : ℤ))

/-- The route-level contraction audit for a combinatorial constant and an
irrelevant scaling factor. -/
def HasRGContraction (C scale : ℝ) : Prop :=
  0 ≤ C ∧ 0 ≤ scale ∧ C * scale < 1

/-- The extended-extraction audit specialized to ‘λ = b^(3-dmax)‘. -/
def HasExtendedExtractionContraction (C : ℝ) (b dmax : ℕ) : Prop :=
  HasRGContraction C (irrelevantScale b dmax)
```

A threshold lemma in the same module records the exact content of the predicate: for $b > 0$ and $d_{\max} \geq 3$, the contraction `HasExtendedExtractionContraction C b dmax` holds if and only if $0 \leq C < b^{d_{\max}-3}$. Thus the entire arithmetic content of the route is the comparison of C with the threshold $b^{d_{\max}-3}$.

2.2 The knife-edge arithmetic

The module machine-checks that the advertised constant contracts at $d_{\max} = 16$ and that the exact one-step gain is $2224 \cdot 2^{-13} = 139/512 < 1/3$:

```
/-- The advertised numerical slack: ‘2224 * 2^-13 < 1‘. -/
theorem rgContraction_2224_two_sixteen :
    HasExtendedExtractionContraction 2224 2 16 := by
  refine ⟨by norm_num, by rw [irrelevantScale_two_sixteen]; norm_num, ?_⟩
  rw [irrelevantScale_two_sixteen]
  norm_num

/-- Exact advertised one-step gain for the extended-extraction constants. -/
theorem rgGain_2224_two_sixteen_eq :
    (2224 : ℝ) * irrelevantScale 2 16 = (139 : ℝ) / 512 := by
  rw [irrelevantScale_two_sixteen]
  norm_num
```

Crucially, the same constant does *not* contract at the nearest even extraction depth below the advertised one. At $d_{\max} = 14$ the threshold is $2^{11} = 2048 < 2224$, so the contraction is arithmetically false; the module proves that any contraction at $d_{\max} = 14$ would force $C < 2048$, and hence that 2224 fails:

```
/-- Any contraction proof at ‘dmax = 14‘ must improve the linear recombination
constant below ‘2048‘.  The manuscript’s advertised bound ‘2224‘ is therefore
insufficient for this lower extraction depth. -/
theorem rgContraction_two_fourteen_forces_constant_lt_2048
    {C : ℝ} (h : HasExtendedExtractionContraction C 2 14) :
    C < 2048 := by
  have hprod : C * irrelevantScale 2 14 < 1 := h.2.2
  rw [irrelevantScale_two_fourteen] at hprod
  nlinarith

/-- Same-constant blocker at the nearest even extraction depth below ‘16‘. -/
theorem not_rgContraction_2224_two_fourteen :
    ¬ HasExtendedExtractionContraction 2224 2 14 := by
  intro h
```

```
  have hC : (2224 : ℝ) < 2048 :=
    rgContraction_two_fourteen_forces_constant_lt_2048 h
  norm_num at hC
```

This is the knife edge: the choice $d_{\max} = 16$ is exactly what makes the advertised constant contract, and the manuscript's own table shows $d_{\max} \in \{10, 12, 14\}$ all fail against $C_1 \leq 2224$ while $d_{\max} = 16$ succeeds. The module `Mettapedia.QuantumTheory.YangMills.RGCrux` packages this into a route-level statement: a mass-gap route that keeps the advertised constant 2224 but uses an even extraction depth strictly below 16 cannot exist, since the required contraction is arithmetically false before any OS / clustering layer is reached.

```
/-- Same-constant lower-extraction certificate in the manuscript's even
canonical-dimension shape: the cutoff is strictly below ‘16‘, is even, and uses
the same advertised recombination bound ‘2224‘. -/
def SameConstantEvenBelowSixteenRGCertificate (dmax : ℕ) : Prop :=
  dmax < 16 ∧ Even dmax ∧ HasExtendedExtractionContraction 2224 2 dmax

/-- No same-constant RG certificate exists for an even manuscript-shape cutoff
strictly below ‘16‘. -/
theorem not_sameConstantEvenBelowSixteenRGCertificate
    {dmax : ℕ} :
    ¬ SameConstantEvenBelowSixteenRGCertificate dmax := by
  intro hcert
  have hdmax : dmax ≤ 14 :=
    even_lt_sixteen_le_fourteen hcert.1 hcert.2.1
  exact (not_rgContraction_2224_two_of_dmax_le_fourteen hdmax) hcert.2.2
```

A strengthened companion theorem, `not_atLeast2048EvenBelowSixteenRGCertificate`, shows that any even lower-extraction repair below $d_{\max} = 16$ would need a genuinely sharper constant ($C < 2048$), not merely the advertised one. We stress that this is an honest delimiter of the arithmetic surface, not a refutation of the route: at the raw natural number $d_{\max} = 15$ the same constant *does* contract ($2^{12} = 4096 > 2224$), and the module records this explicitly so that the obstruction is read precisely as an even canonical-dimension statement matching the manuscript's dimensions $4, 6, \ldots, d_{\max}$.

2.3 Finite strong-coupling spectral gaps and finite OS reconstruction

The module `Mettapedia.QuantumTheory.YangMills.FiniteOSReconstruction` formalizes the finite algebraic core of OS reconstruction: a reflection-positive finite kernel induces a positive pairing on observables, and a transfer operator that is self-adjoint for that pairing descends to the pairing quotient and remains self-adjoint there. On this base, the modules `Z2StrongCouplingGap` and `Z3StrongCouplingGap` build explicit finite single-link abelian lattice gauge models. For $\mathbb{Z}_2$, the one-link heat-kernel transfer at heat time 1 has eigenvalues 1 and $\exp(-1)$, so the dimensionless gap is $-\log(\exp(-1)/1) = 1$:

```
theorem z2StrongCouplingGap_eq_one :
    z2StrongCouplingGap = 1 := by
  simp [z2StrongCouplingGap, z2StrongCouplingLambdaTwo,
    z2StrongCouplingLambdaOne, z2StrongCouplingRatio, Real.log_exp]

/-- End-to-end finite lattice theorem: the explicit ‘Z₂‘ strong-coupling model is
nontrivial, reflection positive, has an OS transfer operator, and has
```

```
dimensionless transfer gap at least ‘1‘. -/
theorem z2StrongCoupling_latticeYangMills_massGap_landmark :
    ∃ _ : Z2StrongCouplingGapCanary,
      (∃ x y : Z2OneLinkConfig, x ≠ y) ∧
        OSReflectionPositive z2StrongCouplingKernel ∧
        z2StrongCouplingGap = 1 ∧
        (1 : ℝ) ≤ z2StrongCouplingGap ∧
        0 < z2StrongCouplingGap := by
  exact
    ⟨z2StrongCouplingPositiveCanary,
      z2OneLinkConfig_nontrivial,
      z2StrongCouplingKernel_reflectionPositive,
      z2StrongCouplingGap_eq_one,
      z2StrongCouplingGap_lower_bound_one,
      z2StrongCouplingGap_positive⟩
```

The $\mathbb{Z}_3$ module proves the analogous landmark with two mean-zero transfer eigenvectors and the same unit gap. These theorems are unconditional and non-vacuous: they exhibit genuine reflection positivity, a genuine finite OS transfer reconstruction, and a strictly positive spectral gap. They are also kept rigorously separate from any continuum claim. Each module carries an explicit scaling diagnostic showing that a finite gap need not survive a naive continuum schedule. For the heat-time schedule $t(a) = a^2$, the *physical* gap (dimensionless gap divided by lattice spacing) closes as $a \to 0$:

```
/-- A concrete gap-closing schedule for the finite model under the heat-time
scaling ‘t(a)=a²‘: the physical gap can be made smaller than any positive
threshold by taking the lattice spacing small enough. -/
theorem z2QuadraticHeatTime_physicalGap_closes :
    ∀ ε : ℝ, 0 < ε →
      ∃ a : ℝ,
        0 < a ∧ a < ε ∧
          z2HeatTimePhysicalGap a (a ^ 2) < ε := by
  intro ε hε
  refine ⟨ε / 2, by linarith, by linarith, ?_⟩
  have hne : ε / 2 ≠ 0 := by linarith
  rw [z2QuadraticHeatTime_physicalGap_eq hne]
  linarith
```

2.4 The continuum endpoint is gated open

The module `Mettapedia.QuantumTheory.YangMills.ProofState` records a status ledger over the route's nodes. The finite RG / extraction surfaces are marked `checked` or `refuted` as appropriate, but the continuum mass-gap endpoint carries an explicit `openGoal` status, gated behind a constructive-QFT promotion barrier (continuum measure, Euclidean covariance, reflection positivity, OS reconstruction, and Hamiltonian mass-gap transfer — none of which is represented for the RG / extraction route, which supplies only a finite ledger):

```
/-- The final mass-gap endpoint remains an open goal in this lane. -/
def yangMillsMassGapEndpointNode : YangMillsProofNode where
  id := "yang-mills.mass-gap-endpoint"
  status := .openGoal
  truthValue := ⟨0, 95⟩
```

```
  evidence := "currentYangMillsMassGapEndpoint_blockedByConstructiveGate blocks promotion
     from the audited RG/extraction surfaces to a continuum mass-gap endpoint ..."
  nextObligation := "Connect the RG and extraction surfaces to a genuine continuum mass-
    gap statement only after the constructive-QFT gate, OS/Hamiltonian transfer
    dependency packet, challenge-world interface packet, and mass-gap promotion gate
    clear."

theorem yangMillsMassGapEndpointNode_open :
    yangMillsMassGapEndpointNode.status = .openGoal := by
  rfl
```

A companion theorem `currentYangMillsMassGapEndpoint_blockedByConstructiveGate` records that the endpoint is open, the constructive checks are all unrepresented, and the only supplied interface is the finite RG / extraction ledger. In short, the formalization does not overclaim: nothing in it asserts a continuum mass gap.

3 Honest status and scope

This lane is, by construction, conservative; we now state precisely what is and is not established.

The contraction constant is an input, not a conclusion. Every arithmetic theorem above takes the recombination constant as a *given real number*. The predicate `HasExtendedExtractionContraction C b dmax` is a statement about C, b, $d_{\max}$ alone; the checked facts `rgContraction_2224_two_sixteen` and `not_rgContraction_2224_two_fourteen` verify the comparison $C \leqslant b^{d_{\max}-3}$ for the specific value $C = 2224$. Nothing in the formalization derives that 2224 (or any $O(1) \cdot b^4$ bound) is the correct value of the manuscript's recombination constant. The Lean ledger is therefore faithful to the route's bookkeeping but agnostic about the route's hardest analytic claim.

The extraction projection: realization succeeds, range-only transfer fails. The sign-only repair is insufficient: even with positive exponent, the displayed binomial/multivariate multiplicities make the corrected finite sum exceed 2 once at least two coordinates are present. We therefore constructed the natural finite polynomial realization rather than treating the omission as a verdict. For a finite Wilson block of actual group-valued links, plaquette holonomies, and a conjugation-invariant Wilson potential, finite polynomials in the plaquette coordinates are gauge invariant. In the explicitly defined weighted coefficient-majorant norm, canonical-dimension truncation has operator constant 1 (module `WilsonBlockMajorant`).

This is our realization of the manuscript's extraction idea, not an attribution to v14. It cannot be transferred to v14's P_{ext} from the specified range alone: `projectionRangeSpecification_does_not_imply_bound_two` constructs an idempotent skew projection onto the same low-coordinate range that violates the bound 2. The positive transfer theorem `coefficientL1_basisJetProjection_le` shows exactly what would suffice: explicit basis and dual-jet matrices whose coefficient-column bounds multiply. The construction attempt is certified in `Dimension16WilsonExtractionCertificate`: F.4's on-shell relation policy and O.9's off-shell policy disagree already on the displayed dimension-6 divergence-square operator; the exact block quotient, lattice representatives, finite basis, dimension-16 dual jets, biorthogonality, and conditioning matrices are not supplied. Appendix I's explicit jets concern the separate observable projection $P^{\mathrm{obs}}_{\leq 4}$. Hence the extraction component is not a standing erratum repair, and C_1 is not derived.

Hard truncation does not descend to the actual trace-word quotient. The third round replaces schematic carriers by the exact object: the SU(2), $b = 2$ Wilson block with normalized Haar measure — 81 vertices, 216 positive links of which 8 are internal, 216 plaquettes, an internal spanning tree with 7 co-tree links, exact boundary-plus-co-tree coordinates, the gauge action, the normalized Wilson potential, and the fixed-boundary product-Haar reference integral with its logarithmic effective block action (modules `SU2WilsonBlockPilot`, `SU2MajorantRepairDecision`). On this object the question "is canonical-dimension truncation even well-defined on the gauge-invariant quotient?" has a negative, intrinsic answer. For any $U \in \mathrm{SU}(2)$ the Cayley–Hamilton identity gives $\mathrm{Tr}(U^2) = \mathrm{Tr}(U)^2 - 2$; writing $a = 2 - \mathrm{Tr}(U)$ and $b = 2 - \mathrm{Tr}(U^2)$ (each of canonical dimension 4), the exact relation $b = 4a - a^2$ yields

$$a^3 b - 4a^4 + a^5 \;=\; 0,$$

whose three monomials have construction-derived canonical dimensions 16, 16, 20. Truncation through dimension 16 deletes only a^5; on the explicit holonomy $\mathrm{diag}(i, -i)$ the truncated expression evaluates to -32, while the exact left-hand side vanishes on every field. Hence the semantic kernel is not stable under representative-wise hard truncation, no function on the evaluated trace-word quotient agrees with that truncation, and the corresponding descended extraction certificate is uninhabited (module `SU2CayleyHamiltonTruncationObstruction`). The witness uses only the Cayley–Hamilton relation — no equations-of-motion quotient and no boundary-extension convention — so it applies to both of v14's relation policies, and it refutes any uniform-in-N architecture that requires representative-wise hard truncation of redundant Wilson trace words.

Corrected constructions do carry norm-below-two extractors. Truncation is, of course, definable on a *chosen basis*, and the third round constructs the two natural repairs exactly. The full associated-graded jet chart on the gauge-fixed block (7 co-tree links, 3 Lie coordinates each, hence 21 coordinates) carries the symmetric-monomial basis of jets through total degree 16, with identity jet matrix and biorthogonal coefficient duals; radial Taylor extraction at radius ratio 1/2 has the exact operator constant $\sum_{d=0}^{16} 2^{-d} \;=\; 131071/65536 \;<\; 2$ (modules `SU2AssociatedGradedExtraction`, `SU2AssociatedGradedExtractionDecision`). Independently, the evaluated one-plaquette trace algebra is exactly a one-variable polynomial algebra: the normal-form theorem $b = 4a - a^2$ (proved semantically complete via an infinite Cayley-rotation family) gives the basis $1, a, a^2, a^3, a^4$ through dimension 16, with extraction constant $\sum_{d=0}^{4} 2^{-d} = 31/16 < 2$ (module `SU2IndependentTraceBasis`). Keeping Appendix O's other factors unchanged, the associated-graded constant yields the conditional ledger value

$$C_{\mathrm{proposed}} \;=\; \tfrac{90832203}{40960} \;\approx\; 2217.583 \;<\; 2224,$$

whose two-source gain at depth 16 is $90832203/335544320 \approx 0.2707 < 1/3$, while depth 15 does not close. This is conditional bookkeeping: the fluctuation, cumulant, and recombination factors remain unproved on the actual map. Two exact theorems keep these constructions honestly separated from v14's object: the full graded chart has dimension 12,875,774,670 and admits no linear equivalence onto any coordinate space of dimension at most 1851, so the manuscript's schematic count requires a genuine invariant quotient that v14 does not construct; and the radial norm used here is a precise repair, not provably the manuscript's intended multivariate norm.

Neither survivor transfers to the canonical (F, D) space. The transfer question is then decided negatively by two independent machine-checked obstructions (module `V14FDQuotientTransferNoGo`, headline `v14_fullFD_quotientTransfer_noGo`). *Derivative blindness:* v14 counts canonical dimension as $2 \cdot \#F + \#D$ (its O.9.1), but radial degree on either chart is blind to derivative factors. The scalar descendant $(\partial^2)^3[(\mathrm{Tr}F^2)^3]$ has canonical dimension 18 yet radial degree 6 on the associated-graded chart and 3 on the trace chart, so both norm-below-two extractors *fix* it — and an explicit two-cell zero-boundary sixth-difference witness evaluates to -6, so the retained operator is nonzero. Both survivors therefore fail to implement the dimension-$\leq$ 16 projection of F.4.2, independently of the choice of quotient basis (`not_associatedRadialImplementsCanonicalCutoff`, `not_independentRadialImplementsCanonicalCutoff`). *Integration-by-parts/block incompatibility:* v14's local norm (F.3) lives on ordinary functions on a finite block retaining boundary data, while both relation branches identify integrated total derivatives with zero. The dimension-6 descendant $\partial^2\mathrm{Tr}(F^2)$ is gauge invariant and evaluates to 1 on an explicit three-plaquette block field (a purely quadratic associated-grade witness gives the same value, so higher Taylor terms are not the cause). An ordinary block-functional realization would have to send one quotient class to both the zero function and a nonzero boundary descendant; the faithful transfer sockets — exact-Wilson, pure associated-graded, and every RG-compatible extension — are therefore uninhabited under both relation policies. Consequently no corrected recursion constant propagates from the constructions above into the bootstrap. The obstruction is structural rather than informational: supplying the omitted dimension-16 coordinates or conditioning matrices cannot make either extractor into the canonical (F, D) projection. A viable replacement needs a derivative-aware filtration whose truncation provably agrees with $2 \cdot \#F + \#D$, together with a boundary-aware complex that retains total-derivative descendants on each block as boundary cochains and imposes integration by parts only at global recombination; constructing exactly that replacement is the current object of this formalization.

The actual-lattice census cannot be Lorentz-only. The derivative-aware boundary-cochain repair makes the filtration and local boundary bookkeeping explicit, but a further obstruction appears before its coordinate basis or dual jets can be supplied. Appendix O.9.3 enumerates only Lorentz-invariant complete contractions. The exact finite-spacing Wilson lattice has only hypercubic symmetry, and the module `SU2LatticeFDCensusNoGo` supplies a gauge-invariant canonical-dimension 8 observable whose fourth Cartan jet is

$$6(F_{01}^4 + F_{02}^4 + F_{03}^4 + F_{12}^4 + F_{13}^4 + F_{23}^4).$$

An explicit rational orthogonal rotation changes its normalized quartic value from 1 to 337/625. Hence no finite span of the $O(4)$-invariant O.9.3 jets contains it. The necessary spanning core, the O.9.2 off-shell certificate, and the F.4.3 on-shell certificate are all formally uninhabited. The witness is derivative-free, so changing EOM, Bianchi, integration-by-parts, trace, Cayley–Hamilton, or boundary-cochain reduction policies cannot repair this same-target symmetry mismatch. The result is already an $N = 2$ counterexample to an architecture asserted uniformly for $\mathrm{SU}(N)$. It is not a no-go for a hypercubic-complete basis or for a controlled continuum projection.

The first hypercubic census succeeds at quartic order and refutes its padded extension. The unpadded carrier of degree-four monomials in the six oriented two-form components has cardinality 126. Exact signed-orbit certificates divide it into four admissible $H(4)$ orbit types — a pure fourth power, adjacent squares, opposite squares, and a rectangle product — plus monomials killed by negative stabilizers. Four signed orbit indicators obey the full hypercubic covariance law, their representative evaluations form the 4×4 identity, and they reconstruct every invariant rational coefficient function. This is a complete result for the derivative-free Cartan quartic restriction, not for the noncommutative (F, D) quotient. The actual gauge-invariant Wilson observable has exact fourth derivative equal to the polynomial realization of coordinate vector $(6, 0, 0, 0)$. Moreover, the curvature observables invariant under every orthogonal spacetime matrix form a proper linear subspace, and this pure-fourth polynomial defines a nonzero class in the quotient by that subspace. Thus the Lorentz-breaking artifact is a coordinate-level theorem, not merely an external rotation counterexample.

The fixed-width full carrier cannot supply that extension faithfully. Inactive field and derivative slots are mathematically padding but remain part of raw equality. The module `V14HypercubicFDCensusPaddingNoGo` constructs two distinct encodings of the empty active syntax and proves `faithful_dimension16_exactCensus_uninhabited`: if every basis column ignores inactive padding, the two basis rows coincide; the required right inverse instead makes them two different rows of the identity matrix. This obstruction precedes Bianchi, EOM, integration-by-parts, trace, and Cayley–Hamilton reduction. It does not show that an $H(4)$ basis cannot exist. It shows that the next construction must first replace the raw carrier by an unpadded/dependent syntax or quotient padding explicitly.

Our unpadded raw character census is exact through dimension 16. OURS. The replacement uses dependent active field and derivative slots, so no inactive coordinate enters equality. An explicit equivalence identifies its canonical-dimension-16 subtype with a separated presentation consisting of an active trace/derivative layout and a rank-16 spacetime-axis assignment; it commutes with the literal hypercubic action and preserves the product of all active tensor signs. The induced linear equivalence identifies the scalar and pseudoscalar covariance submodules on the original syntax with the counted invariant fibers. The exact layout multiplicity is 5,361,612,271,200, and the complete raw carrier has 23,027,949,358,636,282,675,200 elements.

The standard signed four-vector representation of the literal 384-element hypercubic group is constructed over $\mathbb{Q}$. Reynolds-character averaging, with all 24×16 group elements replayed by kernel-checked certificate blocks, gives the scalar multiplicities

$$(1, 0, 1, 0, 4, 0, 31, 0, 379, 0, 5611, 0, 87979, 0, 1400491, 0, 22379179)$$

and pseudoscalar multiplicities

$$(0, 0, 0, 0, 1, 0, 20, 0, 336, 0, 5440, 0, 87296, 0, 1397760, 0, 22368256)$$

at tensor ranks 0 through 16. Since the layout coordinate is inert under $H(4)$, the rank-16 values imply exact raw invariant-space dimensions 119,988,480,745,781,344,800 and 119,929,915,854,943,027,200. No relation is manufactured by the count: field antisymmetry, Bianchi, EOM, integration by parts, trace, and Cayley–Hamilton remain explicit quotient rows. Thus this theorem completes the genuine raw-carrier census, but not the noncommutative physical operator basis or its dual chart.

Our first syntactic quotient removes derivative labels exactly. OURS. The active derivative slots in the raw carrier are labeled, whereas an operator depends only on the ordered derivative word. Interpreting `derivativeOrder` explicitly as the map from ordered positions to active labels separates every raw derivative presentation into a label permutation and the owner–axis datum at each ordered position. Equality of the latter data is proved equivalent to a unique relabeling of the active derivative slots. Consequently the actual quotient of the original exact dimension-16 subtype is linearly represented by ordered derivative words; no canonical representative is assumed.

This alpha quotient commutes with the literal $H(4)$ action and preserves the product of all active tensor signs. Before any physical identity is imposed, the layout multiplicity falls from 5,361,612,271,200 to 5,030,751, and the corresponding carrier has exactly 21,606,911,019,319,296 elements (`HypercubicDimension16AlphaReduction`). Field-label alpha equivalence and the standard field antisymmetry, Bianchi, EOM, integration by parts, covariant-derivative commutator, trace, and SU(2) Cayley–Hamilton identities remain to be imposed. The commutator identity is listed separately because $[D_\mu, D_\nu]X = [F_{\mu\nu}, X]$ is necessary to identify noncommuting derivative words and is not supplied by Bianchi or integration by parts.

Our initial relation rows are equivariant on the same free module. Ours. The signed hypercubic action is extended linearly from the alpha-reduced basis, with the complete active tensor sign attached to each basis vector. Three sparse row families are constructed on this module: uniform field-label renaming, swaps of adjacent ordered derivative positions owned by different field factors, and $F_{\mu\nu} + F_{\nu\mu} = 0$. Each family is proved to intertwine exactly with the signed $H(4)$ action (`HypercubicDimension16RelationEquivariance`). The derivative-shuffle site carries explicit adjacency and different-owner conditions. The adjacency condition prevents a nonlocal transposition from reversing hidden same-owner derivatives. Positions owned by the same field are never identified by a shuffle, since their exchange requires the curvature term in the covariant-derivative commutator. These are equivariance and sparsity theorems; their joint rank has not yet been inferred from the row labels.

Our Bianchi and on-shell EOM rows respect derivative order. Ours. Ordered derivative positions are read from outermost to innermost. A derivative is eligible for Bianchi or EOM reduction only when it is innermost among the positions owned by its field factor. The predicate is preserved by $H(4)$. On such a context the three-term row

$$D_a F_{bc} + D_b F_{ca} + D_c F_{ab}$$

and the contracted on-shell row

$$\sum_{\mu=1}^{4} D_\mu F_{\mu\nu}$$

are constructed explicitly and proved equivariant for the same signed action (`HypercubicDimension16DifferentialRelations`). The EOM rows belong only to the on-shell policy. The innermost condition is essential: moving either identity through deeper derivatives without curvature commutators would change the represented operator. No rank or completeness claim is deduced merely from these formulas.

Our integration-by-parts row is an integrated action identity. Ours. A globally outermost derivative position is chosen and its owner is varied over every field factor. The sum of those basis rows is the product-rule expansion of an integrated total covariant derivative and is proved equivariant under the signed $H(4)$ action. The outermost condition is preserved by the action and is compatible with the different-owner shuffle rows. This is an action-sector integration-by-parts relation; it is not asserted as pointwise vanishing of a local derivative.

The same module supplies the elementary traceless-letter row: a fixed point of the trace permutation represents a one-letter trace and is killed because every covariantly differentiated field strength is $\mathfrak{su}(2)$-valued and traceless. Disjoint cycles already encode cyclic words without a chosen starting point or ordering of trace factors, while uniform field-label alpha rows identify conjugate labeled presentations (`HypercubicDimension16IBPTraceRelations`). Higher SU(2) trace identities remain separate.

Our covariant-derivative commutator is an exact cross-sector row. OURS. A site consists of two consecutive positions in the ordered derivative word of a single field letter; derivatives belonging to other field letters may occur between them in the global word. The construction removes precisely those two positions while retaining the relative order of their complement, adds a curvature letter with the two derivative axes, and inserts it immediately before or after the selected field in the same trace cycle. The resulting four-term row is

$$D_a D_b X - D_b D_a X - F_{ab} X + X F_{ab} = 0.$$

Both curvature insertions have the same canonical dimension as the two derivative terms. The trace-cycle insertion equations, dimension identity, complement-order embedding, active tensor sign, and signed $H(4)$ intertwining law are proved exactly (`HypercubicDimension16CovariantCommutator`). Thus no same-field derivative swap is imposed without its curvature commutator. This theorem constructs the relation operator; it does not infer its rank or the rank of the combined quotient. The higher SU(2) trace identities are supplied next as independent row schemas.

A literal seven-field, two-derivative instance proves that this cross-sector coupling survives both exact contextual normalization and simultaneous field relabeling. Its two eight-field curvature insertions are identified by independent fixed-width witnesses with two distinct entries of the checked positive orbit chart. The eight-field part of the corresponding physical-relation column is exactly

$$-e_{10513} + e_{10515},$$

and its signed Reynolds average is nonzero by off-diagonal orbit separation and the nonzero diagonal coefficient theorem. Hence the descended physical relation operator is not block diagonal in field count: the derivative-free eight-field trace quotient cannot replace the joint dimension-sixteen physical quotient without the incoming commutator block (`HypercubicDimension16CovariantCompressionWall`).

Our joint invariant quotient is fixed before rank computation. OURS. Let V be the genuine simultaneous-field-relabel orbit module, $I = V^{H(4)}$ its signed hypercubic fixed submodule, and $R_p \subset V$ the range of the joint physical-relation operator under policy p. We define the invariant relation submodule canonically as

$$J_p = \iota^{-1}(R_p) \subset I,$$

where $\iota : I \hookrightarrow V$ is the inclusion, and define the scalar physical quotient to be I/J_p. Membership in J_p is therefore exactly membership of the underlying invariant vector in the full relation range; the structural definition materializes neither a Reynolds matrix nor a dense relation matrix. The off-shell generator space embeds in the on-shell one without adding an off-shell EOM row, so $J_{\text{off}} \subseteq J_{\text{on}}$ and the on-shell quotient dimension is no larger. This fixes the honest coordinate-free target for sparse certificates; it does not infer either quotient dimension or conditioning (`HypercubicDimension16JointPhysicalQuotient`).

Our SU(2) trace rows include genuine multitrace rewiring. Ours. For traceless 2×2 matrices the polarized Cayley–Hamilton identity

$$AB + BA = \mathrm{Tr}(AB)I$$

is proved over an arbitrary commutative ring, together with its trace-context form

$$\mathrm{Tr}(UABV) + \mathrm{Tr}(UBAV) = \mathrm{Tr}(AB)\mathrm{Tr}(UV).$$

On the finite carrier, the right-hand side is constructed by splitting the adjacent A, B letters into a two-cycle and reconnecting the remaining trace word. The three changed permutation edges are proved explicitly.

Adjacent splitting alone does not visibly generate identities which join several trace cycles. Accordingly, for every injection of three cut labels we also construct the standard six-term fundamental 2×2 trace row

$$\mathrm{Tr}A\mathrm{Tr}B\mathrm{Tr}C - \mathrm{Tr}(AB)\mathrm{Tr}C - \mathrm{Tr}(AC)\mathrm{Tr}B - \mathrm{Tr}(BC)\mathrm{Tr}A + \mathrm{Tr}(ABC) + \mathrm{Tr}(ACB) = 0.$$

The identity is proved by matrix evaluation, while its carrier realization is the signed sum over the identity, three transpositions, and two three-cycles acting on the selected outgoing trace edges. Thus its terms honestly join and split cyclic and multitrace layouts. Both row families preserve every axis assignment and active tensor sign and are signed-$H(4)$ equivariant (`HypercubicDimension16SU2TraceRelations`). These are relation and equivariance theorems; no joint rank or quotient conditioning is inferred from the schemas alone.

Our sparse trace-rank replay is kernel checked, and its rational row arithmetic has an exact semantics. Ours. A bounded exact enumeration on the genuine eight-field scalar orbit carrier selects 11,556 integer trace-relation rows against the 11,654 orbit coordinates, leaving 98 candidate free columns. The selected rows contain 25,379 nonzero integer entries. Exact elimination produces 27,568 nonzero rational entries, uses only 675 reductions by previous pivots in total, and uses at most 25 such reductions for any one row. The rational replay is split into forty-five 256-row certificates and one 36-row certificate. Lean's kernel checks every reduction and every normalized row using explicit numerator–denominator arithmetic; no dense matrix is materialized and no evaluator axiom is used.

An independent semantic layer now interprets every valid integer pair (n, d) as $n/d \in \mathbb{Q}$. It proves that exact zero tests and cross-multiplication equality are sound, that sparse insertion and cancellation denote vector addition, that the complete fold used for a scaled row subtraction denotes the corresponding subtraction in the rational coordinate module, and that pivot normalization denotes inverse scalar multiplication. Validity is preserved by every one of these operations under the explicit nonzero-denominator hypotheses (`HypercubicDimension16TraceRankSemantics`).

The forty-six checked ranges are now assembled into one typed witness for every selected row. The normalized rows have pairwise distinct maximal pivots and are therefore linearly independent. Strong induction through the exact replay proves that every normalized row lies in the span of the selected raw rows. Since both indexed families have 11,556 members, the selected raw rows themselves are linearly independent over $\mathbb{Q}$, and their rational span has finrank exactly 11,556 (`HypercubicDimension16TraceRankTheorem`).

We also reran the sparse elimination independently modulo the two primes 1,000,000,007 and 1,000,000,009. Each bounded run examined 5,455,784 generated candidate rows and produced 11,556 pivots with 27,568 nonzero normalized-basis entries. The prime-specific coefficients and reduction traces are stored separately in canonical increasing-support order. Lean proves that both moduli are prime and distinct, checks the stored basis and trace shapes, and replays every one of the 11,556 selected integer rows in forty-six chunks at each prime (`HypercubicDimension16TraceRankModularCertificate`). These two prime-field replays are independent arithmetic audits. The exact $\mathbb{Q}$ theorem above remains the semantic rank result, and the modular checks do not by themselves certify exhaustive generation of the physical relation operator.

The independently defined target does not use the encoded coordinate table. On the genuine eight-field field-relabel orbit module, we take the span of the literal signed-orbit rows $v - hv$. This span is proved to be exactly the kernel of the Reynolds projector: the forward inclusion uses right invariance of the group average, while the reverse inclusion writes $v - Pv$ as the average of the rows $v - hv$. Consequently the signed coinvariant quotient is linearly equivalent to the previously constructed Reynolds-invariant subspace and has finrank exactly 11,654 (`HypercubicDimension16SignedOrbitCoinvariants`). The encoded orbit chart is proved below to coordinatize this independently defined standard quotient; it does not define the quotient by its own table.

We also constructed independent one-way invariants for the encoded chart. For each field label the first records the eight successive oriented-plane colors around the literal trace permutation, together with all eight return-to-start bits, as a base-twelve rooted code. Both the sorted multiset of the eight codes and the standard symmetric polynomial

$$\prod_{f=1}^{8}(1 + c_f)$$

are proved invariant under simultaneous field relabeling before they are descended to the genuine orbit carrier. Reflections do not change the plane transport, so minimization requires only the 24 unsigned coordinate permutations; the resulting minimum is proved invariant under the full literal hypercubic group (`HypercubicDimension16TraceOrbitSignature`). Distinct minima therefore certify distinct genuine carrier orbits, but no converse or completeness property is assumed.

The fixed-width payload contains 11,654 product-signature minima, all distinct. It stores the values in representative order and strictly sorted order together with mutually inverse rank tables. The kernel replay is partitioned into 182 ranges of 64 representatives and one final range of 6; each range recomputes all 24 unsigned images directly from the checked trace permutation and plane colors. The global certificate separately checks the rank inverse and the sorted table in eleven 1024-entry blocks and one 390-entry tail; the same bounded replay checks all 11,653 adjacent strict inequalities before using the one-way orbit theorem (`HypercubicDimension16TraceOrbitProductSignatureCertificate`). Exhaustivity of the chart remains a separate obligation.

For the stabilizer sign, an even coarser standard invariant suffices. The six plane multiplicities are defined on the labeled carrier, proved invariant under field relabeling, and descended to the genuine orbit carrier. Exact enumeration gives 32,940 unsigned plane-histogram stabilizer pairs and no negative tensor sign; every representative also has even incidence at each coordinate axis. The two-form tensor sign factors exactly into its unsigned permutation part and its pure-reflection part. Accordingly, the bounded certificate checks the 16 pure reflections and the 24 unsigned plane-histogram implications separately for each representative; the factorization then supplies all 384 signed elements. This tests only a necessary condition for an actual stabilizer and assumes completeness of neither the plane histogram, the rooted code, nor the product signature (`HypercubicDimension16TraceOrbitPlaneHistogram`, `HypercubicDimension16TraceOrbitHistogramSignCertificate`).

For each decoded target, take the Reynolds average of its standard coordinate vector. Pairwise orbit separation makes the evaluation matrix of these vectors diagonal, and positive stabilizers make every diagonal entry a strictly positive rational number. The 11,654 Reynolds vectors are therefore linearly independent. Their cardinality equals the independently proved finrank 11,654 of the invariant subspace, so they form a basis; transport through the Reynolds equivalence gives a basis of the standard signed coinvariant quotient (`HypercubicDimension16TraceOrbitBasis`). This proves that the chart coordinatizes the independently defined quotient rather than defining it by enumeration.

Every selected raw row is now proved to have support inside the finite 11,654-coordinate ambient module. Restriction to that module preserves linear independence, so rank–nullity gives dimension 98 for the quotient by the selected-row span. The 11,556 normalized rows together with the 98 certified free coordinate vectors form a basis of the ambient module. Their free images form a basis of the selected-row quotient, and the corresponding 98×98 analysis matrix is exactly the identity (`HypercubicDimension16TraceFiniteQuotient`).

Every selected encoded row is also identified with an actual Reynolds image of a literal standard trace row. The certificate decodes the stored source term, field relabeling, hypercubic transport, tensor sign, and integer coefficient of each nonzero coordinate. Thirty ranges of 256 witnesses cover the 7680 one-letter traceless rows; sixty ranges of 64 witnesses and one range of 36 cover the 3876 six-term fundamental rows. A separate finite coefficient replay is proved equal to the corresponding semantic finitely supported vectors, so the checked payload cannot introduce a new relation by its encoding. Consequently the 11,556-dimensional selected span is a subspace of the genuine standard trace-relation span (`HypercubicDimension16PhysicalTraceWitnessBridge`).

Independently of that finite selection, the genuine eight-field orbit module carries the three literal standard SU(2) trace schemas: one-letter tracelessness, the polarized Cayley–Hamilton anticommutator row, and the six-term fundamental 2×2 trace row. Their signed $H(4)$ intertwining law is proved before any rank computation. Each of the 98 rational plane-matrix assignments is kernel-checked to be traceless, and ordinary product-of-traces evaluation annihilates every row in all three families, including after arbitrary hypercubic transport (`HypercubicDimension16FieldEightPhysicalTrace`, `HypercubicDimension16PhysicalTraceEvaluationInvariant`). Their full 98×98 evaluation matrix is checked together with an exact rational inverse. The resulting 98 annihilation functionals are independent on the certified free classes, giving the opposite rank inequality to the selected-row inclusion. Therefore the complete literal standard trace-relation submodule has finrank 11,556, its derivative-free eight-field quotient has finrank exactly 98, and the certified free classes form an honest quotient basis (`HypercubicDimension16PhysicalTraceEvaluationCertificate`, `HypercubicDimension16PhysicalTraceQuotient`).

The genuine 384-element Reynolds normalization makes the public evaluation matrix equal to the stored unsigned matrix divided by 24. Its exact dual is consequently 24 times the stored inverse. These duals are proved biorthogonal to the 98 free classes and reconstruct every quotient vector. The sharp coefficient-sup analysis constant is

$$\frac{14739388451343}{30664304},$$

attained by row 30 (`HypercubicDimension16PhysicalTracePublicConditioning`). If the previously isolated fluctuation, cumulant, and recombination hypotheses are transported unchanged to this evaluation-sup norm, the conditioned ledger is

$$\frac{928581472434609}{1742290}.$$

At block factor two its least linear contraction depth is 32, and its least two-source closing depth is 34 (`HypercubicDimension16TraceEvaluationBootstrap`). This last arithmetic is conditional and concerns only the derivative-free trace sector; it supplies neither the derivative-coupled joint quotient nor an interacting Wilson-RG estimate.

Our quantitative restoration theorem is tree-level and constant-Cartan. OURS. The derivative-free Cartan octic restriction contains 1287 monomials and exactly 17 signed $H(4)$ orbit coordinates. Representative evaluation supplies exact duals and the identity conditioning matrix, hence analysis and synthesis both have coefficient-sup factor one. For the actual SU(2) Cartan link field, the normalized Wilson eighth jet is

$$-2a^{12}\sum_{\mu<\nu}F_{\mu\nu}^{8},$$

so division by 8! leaves only the pure-octic coordinate, equal to $-a^{12}/20160$. If $|F_{\mu\nu}|\le M$ and $|a^2F_{\mu\nu}|\le 1$, the actual six-plaquette density obeys

$$\left|\mathcal{W}_a(F)-\sum_{\mu<\nu}F_{\mu\nu}^{2}\right|\le\frac{5}{8}a^4M^4,$$

and therefore converges to the $O(4)$-invariant quadratic density along every eventually positive lattice-spacing net tending to zero. The corresponding two-block rotation error is at most $(5/2)a^4M^4$.

The 17 coordinates lift to the postponed-integration-by-parts boundary complex; shared-boundary coboundaries cancel under global recombination, and the degree-zero quotient retains rank 17. The newly derived extraction factor is exactly one. Keeping the fluctuation, cumulant, and recombination bounds as explicit hypotheses gives the conditional recursion budget 5544/5; the two-source bootstrap then closes at depth 15 and fails at every depth at most 14. None of these statements constructs an interacting continuum measure, proves uniform Symanzik estimates for quantum correlators, verifies Osterwalder–Schrader reconstruction, or proves a mass gap.

The finite gaps are real but Clay-irrelevant. The $\mathbb{Z}_2$ and $\mathbb{Z}_3$ spectral gaps are honest theorems, but they are abelian, single-link, strong-coupling, and finite. They exemplify the finite OS-reconstruction mechanism; they are not lattice approximations to a continuum SU(N) theory, and the accompanying scaling diagnostics show that the physical gap can close under a naive continuum schedule. They establish the machinery, not the target.

The continuum mass gap is correctly open. The continuum endpoint is gated `openGoal` with no claimed support, and the constructive-QFT prerequisites are all marked unrepresented. The formalization makes no claim, in either direction, about the existence of a continuum Yang–Mills mass gap.

4 What remains, exactly

Machine-checked since the first version of this report

A second formalization round closed several items that the first version listed as open, and sharpened others into theorems:

- **The two-source bootstrap threshold.** Linear contraction ($\Lambda < 1$) is *not* sufficient for the manuscript's bootstrap; the scalar induction closes iff $a = C\, b^{3-d_{\max}} \leq 1/3$ (`scalar_bootstrap_two_sources`). At $C = 2224$, $b = 2$: depth 14 fails even linear contraction, depth 15 contracts linearly but the bootstrap *fails* (`not_twoSourceBootstrapSlack_2224_two_fifteen`), depth 16 closes. The true reason for $d_{\max} = 16$ is the 1/3 threshold, not parity of the depth — and the threshold is *dynamical*: an explicit nonlinear toy orbit realizes the recursion with equality and exits the ε^*-ball in one step at depth 15 (`toyRG_ball_escape_two_fifteen`, `benToyRGBootstrapRealization`).

- **The extraction realization and transfer obstruction.** The manuscript's literal displayed sum is ≥ 3 for every $n \geq 1$ and never ≤ 2 (`three_le_literalExtractionOperatorSum`, `not_literalExtractionOperatorSum_le_two`). On an explicit finite Wilson-block polynomial algebra, coefficient-majorant truncation is norm one and gauge invariant (`WilsonBlockMajorant`). A matrix column-sum theorem transfers through explicit basis and dual jets; v14 does not provide these for $P_{\text{ext}} \leq 16$, and a skew-projection counterexample proves that no bound follows from the range alone (`ExtendedExtractionTransfer`). The source-certificate audit further proves that F.4 and O.9 select different EOM relation policies and records the absent lattice representatives, quotient basis, dimension-16 jets, biorthogonality, and conditioning rows (`Dimension16WilsonExtractionCertificate`).

- **The support-growth dichotomy.** The two collar-growth statements the manuscript makes are inequivalent, and both branches are now tight theorems: a coarsen-then-thicken recurrence $r_{j+1} \leq r_j/b + A$ yields the uniform bound $r_0/b^J + Ab/(b-1)$, while a thicken-only recurrence $r_{j+1} \leq r_j + A$ yields exactly $r_0 + AJ$ (`support_growth_dichotomy`).

- **Gap $\Leftrightarrow$ clustering, finite class.** Reflection positivity, self-adjointness, and a q-contraction on the vacuum complement give exponential decay of truncated two-point functions at rate $-\log q$ (`abs_truncatedTwoPoint_le_exp`); conversely, RP plus per-vector moment decay $\langle v, T^n v\rangle \leq C_v q^n$ on a dense subset gives the gap (`eigenvalue_abs_le_of_moment_decay`) — note the *per-vector* constants: only the rate must be uniform, which weakens what the manuscript's Assumptions K.7/M.5 need to supply. A regression records that an additive t-independent tail defeats the conclusion (`moment_decay_additive_defeat`), so large-field remainders must decay, not merely be small.

- **A non-abelian instance.** The quaternion group $Q_8 \subset SU(2)$ single-link model: machine-checked non-commutativity ($ij = k \neq -k = ji$), reflection positivity, spectral gap carried by the four-dimensional spin isotypic component, and clustering; together with the $\mathbb{Z}_2$ instance this discharges the lattice-side inputs (`massGap`, `reflectionPositive`, `clustering`) of the finite continuum-gate conditional (`z2_continuumMassGap_of_reconstructionMachine`).

Machine-checked in the third round

A third round moved from schematic carriers to the exact SU(2), $b = 2$ Wilson block (§3) and decided the extraction stage on it:

- **Hard truncation refuted intrinsically.** The Cayley–Hamilton relation $a^3b - 4a^4 + a^5 = 0$, with monomial dimensions $(16, 16, 20)$ and the explicit -32 evaluation, proves representative-wise canonical truncation is not well-defined on the evaluated trace-word quotient (`SU2CayleyHamiltonTruncationObstruction`).

- **Exact norm-below-two extractors constructed.** The 21-coordinate associated-graded jet chart has extraction constant 131071/65536 and the one-plaquette trace quotient has 31/16, both certified with explicit bases, identity jet matrices, and biorthogonal duals; the conditional ledger value $90832203/40960 \approx 2217.6$ closes the two-source bootstrap at depth 16 and fails at depth 15 (`SU2AssociatedGradedExtraction`, `SU2IndependentTraceBasis`).

- **A certified dimension separation.** The full graded chart has dimension 12,875,774,670 and admits no linear equivalence onto any space of dimension $\leq$ 1851, so v14's schematic operator count requires an invariant quotient the manuscript does not construct (`SU2AssociatedGradedExtractionDecision`).

- **The (F, D) transfer socket is uninhabited as written.** The dimension-18/radial-degree-6 descendant and the integration-by-parts/block witness $\partial^2 \mathrm{Tr}(F^2)$ jointly refute the faithful transfer sockets for both relation policies and all RG-compatible extensions (`v14_fullFD_quotientTransfer_noGo`).

Machine-checked in the fourth round

A fourth round constructed the derivative-aware boundary-cochain scaffold and then decided whether its dimension-16 coordinate socket could be filled by the manuscript's Lorentz-only operator census:

- **The repaired filtration and boundary bookkeeping are mechanical.** The canonical-dimension filtration, boundary-aware cochain coordinates, coefficient projection, and conditional scalar bootstrap are machine-checked. Their analytic Wilson-RG factors remain explicit inputs.

- **The actual-lattice O.9 census is refuted at dimension 8.** The exact SU(2) Wilson observable, Cartan chart, fourth-jet calculation, and rational rotation witness prove that the hypercubic quartic jet is not Lorentz invariant (`SU2LatticeFDCensusNoGo`).

- **Both relation branches and six same-target variants are closed.** Neither the off-shell O.9.2 policy nor the on-shell F.4.3 policy admits a complete Lorentz-only certificate. EOM, Bianchi/IBP, cyclic/multitrace, SU(2) Cayley–Hamilton, and postponed-IBP variants share the same uninhabited spanning core.

- **The live fork is explicit.** One must either enumerate an $H(4)$-complete finite-spacing basis and rederive its duals and constants, or prove a Symanzik/continuum projection with quantitative decay of the Lorentz-breaking artifact coordinates. No numerical factor transfers automatically from the refuted census.

Machine-checked in the fifth round

The first branch of the live fork was pursued directly:

- **The exact hypercubic group and finite raw syntax are mechanical.** Signed coordinate permutations form a concrete group of order 384. The raw surface records field and derivative indices, trace cycles, derivative ownership/order, canonical dimension through 16, scalar/pseudoscalar characters, and separate relation-family labels. A label still supplies no relation row by itself (`V14HypercubicFDCensus`).

- **The unpadded dimension-8 quartic census is complete.** The 126 Cartan quartic monomials have checked signed-orbit certificates. Four surviving orbit functions form a basis, representative evaluation supplies exact duals, and the conditioning matrix is the identity (`V14HypercubicQuarticCensusCertificate`, `V14HypercubicQuarticBasis`).
- **The actual Wilson and artifact coordinates are certified.** The Wilson fourth jet is exactly coordinate vector $(6, 0, 0, 0)$. The $O(4)$-invariant function subspace is proper, and the pure-fourth polynomial is nonzero in its quotient (`V14HypercubicQuarticWilsonBridge`).
- **The padded dimension-16 extension is refuted.** Two distinct fixed-width encodings of the empty active syntax differ only in inactive padding. Any active-syntax-faithful basis gives them equal rows, which is incompatible with the right-inverse identity required by `ExactCensusCertificate` (`V14HypercubicFDCensusPaddingNoGo`).
- **The corrected carrier is specified without padding.** The obstruction identifies finite unpadded/dependent (F, D) syntax, or a checked padding quotient, as the required carrier on which the $H(4)$ action and relation rows must descend. No dimension-8 constant transfers automatically. The unpadded construction and its raw character census are supplied in the results below; the relation quotient remains separate.

Machine-checked on the unpadded dimension-16 carrier

All constructions in this subsection are ours and are independent of any claim that the manuscript already contains the needed census.

- **The counted carrier is the actual active syntax.** Exact dimension-16 shapes are classified by $f \in \{0, \ldots, 8\}$ with $d = 16 - 2f$. A checked equivalence identifies the corresponding subtype of the unpadded raw (F, D) syntax with the separated layout–rank-16 tensor presentation, preserves the literal action and tensor sign, and induces a linear equivalence on the covariance submodules. Its exact cardinality is 23,027,949,358,636,282,675,200 (`HypercubicUnpaddedFDSyntax`, `HypercubicRawFDDimension16Census`).
- **Derivative-label alpha equivalence is quotiented exactly.** The raw derivative order is separated into a disposable active-label permutation and the owner–axis word indexed by ordered positions. Equality of ordered words is equivalent to a unique relabeling; the quotient of the original exact carrier is explicitly equivalent to the reduced carrier, equivariant under $H(4)$, and tensor-sign preserving. Its exact layout and carrier counts are 5,030,751 and 21,606,911,019,319,296 (`HypercubicDimension16AlphaReduction`).
- **The first sparse relation operators are equivariant.** Uniform field-label renaming, adjacent different-owner derivative shuffles, and field-strength antisymmetry are explicit rational rows on the same free module. Their signed $H(4)$ intertwining laws are proved before any rank claim; same-owner derivative swaps are excluded (`HypercubicDimension16RelationEquivariance`).
- **The standard rows form one equivariant physical-relation operator.** Our off-shell policy combines antisymmetry, innermost Bianchi, globally outermost integrated IBP, traceless one-cycles, the covariant commutator, and the two ordinary SU(2) Cayley–Hamilton trace consequences; our on-shell policy additionally enables innermost EOM. The signed source and target actions are rational $H(4)$ representations, the joint operator intertwines them, and

its image is invariant. The same statements hold after the exact antisymmetry/derivative-localization quotient. This is an equivariance and assembly theorem, not yet a physical-rank claim (`HypercubicDimension16PhysicalRelationOperator`).

- **The physical-relation operator reaches the genuine field-label orbit carrier.** Our ordinary simultaneous field-permutation quotient of the oriented local carrier is lifted to a rational linear map. Explicit simultaneous-renaming difference rows span exactly its kernel, and the induced quotient is linearly equivalent to the orbit module. Its signed $H(4)$ action is a representation, the quotient map intertwines the contextual and orbit actions, and composing it with the joint physical-relation operator leaves an invariant image. Thus the operator target is now the same literal finite orbit carrier whose character was counted above; no sectorwise rank or conditioning is inferred here (`HypercubicDimension16PhysicalOrbitOperator`).

- **The descended operator has an exact sparse field-sector band.** Our source-field function is defined on every standard relation generator. Every raw row is supported in that field sector, except that the two curvature-insertion terms of the covariant commutator lie in the successor sector. Explicit rational field filters commute with both the contextual normalization and the simultaneous field-relabel quotient. Consequently each column of the physical operator on the genuine orbit carrier is supported in its source sector and, when present, its immediate successor. This is a block upper-bidiagonal incidence theorem, not a rank or conditioning result (`HypercubicDimension16PhysicalRelationBlocks`).

- **The honest eight-field scalar invariant space has dimension 11,654.** Our signed permutation representation is defined directly on genuine simultaneous-field-relabel orbits. Its 384-term Reynolds average is proved to be the projector onto the fixed subspace, and the trace of each action is the literal signed fixed-orbit character. The exact conjugacy-compression identity therefore gives scalar invariant finrank 11,654 in the $(f, d) = (8, 0)$ sector, without choosing orbit representatives or inserting a padded carrier (`HypercubicDimension16PhysicalInvariantSpace`).

- **Bianchi and EOM preserve noncommuting derivative order.** The cyclic Bianchi row and the contracted on-shell EOM row are supplied only at an innermost derivative of the selected field word. Their signed $H(4)$ intertwining laws are checked, and EOM remains disabled in the off-shell policy (`HypercubicDimension16DifferentialRelations`).

- **Integrated IBP and traceless one-cycles are explicit.** The total-derivative product-rule row is supplied at a globally outermost derivative position, and a one-cycle of the trace permutation is killed by tracelessness. Both row families are signed-$H(4)$ equivariant (`HypercubicDimension16IBPTraceRelations`).

- **The covariant-derivative commutator crosses sectors honestly.** Two consecutive derivatives of one field word are removed in complement order, an F_{ab} letter is inserted on either side of that field in its trace cycle, and the four-term commutator row preserves canonical dimension and active tensor sign. Its signed-$H(4)$ intertwining law is checked. For an explicit seven-field, two-derivative column, the two successor terms occupy distinct checked orbit coordinates, its exact eight-field component is nonzero, and its Reynolds projection remains nonzero. Thus the physical operator is provably not block diagonal in field count (`HypercubicDimension16CovariantCommutator`, `HypercubicDimension16CovariantCompressionWall`).

- **The** SU(2) **trace identities rewire the actual trace permutation.** The traceless anticommutator row splits an adjacent pair into a two-cycle and reconnects its context. The six-term fundamental 2×2 trace row additionally joins and splits distinct cycles through three selected edges. Both semantic matrix identities and both signed $H(4)$ intertwining laws are checked (`HypercubicDimension16SU2TraceRelations`).

- **The derivative-free eight-field standard trace quotient is complete.** The independently checked orbit chart has 11,654 coordinates. A literal 11,556-row standard trace subfamily is linearly independent, while 98 ordinary traceless-matrix evaluations annihilate every standard trace row and have an exact rational inverse on the free classes. Hence the complete standard trace submodule has finrank 11,556 and its quotient has finrank 98. The free classes form a basis with explicit public duals. In genuine Reynolds normalization the sharp row-ℓ^1 dual constant is 14739388451343/30664304, attained at row 30. This is the derivative-free trace quotient, not the derivative-coupled joint physical quotient (`HypercubicDimension16PhysicalTraceQuotient`, `HypercubicDimension16PhysicalTracePublicConditioning`).

- **The first honest cycle-index compression target remains enormous.** For our explicitly defined cycle-index expression—oriented two-form letters, one ordered local derivative word per field, and simultaneous unlabeled field/trace-cycle data—the eight field sectors have identity specialization 47,231,161,854. Its scalar and pseudoscalar $H(4)$ Reynolds values are 246,136,792 and 246,121,092. The literal 384-element action variable is reduced to fourteen power-trace classes only after a checked axis-recurrence/group-power bridge, all traces through exponent 16, determinant parity, class assignments and multiplicities, the cycle-letter recursion, and the final Reynolds-average equality. These numbers impose none of Bianchi, EOM, IBP, the covariant commutator, or the matrix trace identities. A concrete finite labeled carrier now contains exactly a trace permutation and, on each field label, an oriented two-form plus a finite ordered derivative-axis word of the prescribed total degree. Genuine field-relabel and $H(4)$ actions on this carrier commute; the tensor-sign cocycle is field-relabel invariant, descends to the orbit quotient, and retains its group law. The remaining gap is a character theorem identifying that orbit carrier with the displayed cycle-index expression. For the derivative part, stable projection of a global owner–axis word to one ordered axis word per owner is now surjective with a canonical owner-by-owner right inverse; its semantic quotient is exactly the finite local-word family, commutes with field relabeling and $H(4)$, and every already-defined adjacent different-owner shuffle row lies in its kernel. A head-matching induction now proves the converse without counting: if two tagged global words have the same owner-local words, the head occurrence can be moved left through only different-owner entries, after which the proof recurses. Consequently the equivalence closure of the literal adjacent generators is exactly the semantic kernel, and the generated quotient is equivalent to the finite local-word family. This result is now also linearized over the rationals: the kernel of the free-module localization map is exactly the span of the literal two-term adjacent-shuffle rows, and its quotient is linearly equivalent to the free module on local-word families. At the single field-strength-letter level, the ordered-pair space now has an explicit $16 = 6 + 10$ linear decomposition: the six coordinates are the oriented two-planes, while four diagonal and six symmetric coordinates synthesize exactly the span of the literal $F_{ab} + F_{ba}$ rows. Hence the normalization kernel is precisely that row span, the quotient is the six-plane space, and the normalization intertwines the signed $H(4)$ actions. More generally, for every finite family of field slots the product normalization has

kernel exactly the span of the literal slotwise antisymmetry rows, with an explicit quotient equivalence to oriented-plane assignments. The antisymmetry and linear shuffle theorems are now assembled while retaining the actual trace layout and then lifted through the dependent dimension-16 union. On that exact carrier, the combined kernel is precisely the span of the already-defined position-indexed antisymmetry and adjacent-shuffle rows; its quotient is the free module on trace layouts, oriented-plane assignments, and owner-local derivative words. The combined map also intertwines the literal signed $H(4)$ actions. That exact fixed-sector target is now explicitly equivalent to the finite labeled local-word carrier: finite derivative functions and owner-local lists are related by inverse maps, the trace permutation and oriented planes are preserved, and the equivalence intertwines field relabeling and the signed $H(4)$ action, including its tensor sign. It therefore descends to an explicit equivalence of the genuine field-label orbit quotients. The remaining character is now defined independently as the tensor-sign-weighted sum over fixed points of the literal $H(4)$ action on that orbit carrier. A general weighted twisted Burnside theorem, proved by transporter counting and orbit–stabilizer, specializes here to show that the labeled twisted fixed-point numerator is exactly $f!$ times this actual orbit character. The first local factor of the remaining finite cycle theorem is also separated semantically: the signed fixed-point sum on one exact local field letter (one oriented plane and an ordered derivative word of length k) is proved to be $\chi_{\Lambda^2 V}(h^r)\chi_V(h^r)^k$ for a relabeling cycle of length r. The exact derivative-degree convolution is now separated as well. Our dependent cycle-seed carrier chooses one exact local letter for each relabeling cycle, lets a cycle of length r act by h^r, and enforces $\sum rk = d$ through exact residual-degree fibers. Its direct signed fixed-point sum is proved equal to the recursive partition derivative character. The literal one-cycle fixed equation is now bridged as well: for every positive r, a family of actual local letters satisfying the one-step relabeling/h equation is explicitly equivalent to one seed fixed by h^r, and the product of its r one-step tensor signs is proved to be the tensor sign of h^r on that seed. These literal families are now assembled over every positive cycle list in a second dependent carrier: zero-length cycles are uninhabited, exact residual derivative degree is retained, and its complete signed weight sum is proved equal to both the cycle-seed character and the recursive partition character. Our concrete disjoint-cycle carrier takes the recursive sum of the finite blocks displayed by that list and rotates each block by one step. Its literal fixed-decoration subtype is explicitly equivalent to the budget-stable dependent carrier, with exact preservation of the full tensor-sign product. Consequently its signed fixed-decoration character is exactly the recursive partition character. The conjugacy step is now closed as well. The displayed positive partitions through eight fields are proved to be exactly the standard integer partitions; each has a concrete disjoint-cycle representative with the advertised cycle partition. The fiber of the standard partition map on actual field permutations is explicitly equivalent to the corresponding ordinary conjugacy class. The decoration character and centralizer cardinality are constant on that fiber, and the standard class-cardinality times centralizer-cardinality identity cancels the field factorial over $\mathbb{Q}$. Hence the actual signed orbit character equals the recursive cycle-partition character for every $H(4)$ element and every nonempty field sector through eight. The scalar and pseudoscalar values 246,136,792 and 246,121,092 are therefore exact invariant dimensions of this intermediate quotient. They impose none of the remaining physical relations and are neither a physical-basis count nor a compression wall (`HypercubicDimension16CompressionCensus`, `HypercubicDimension16CompressionReplayCertificate`, `HypercubicDimension16CompressionSectorCe`
`HypercubicDimension16CompressionSemanticBridge`, `HypercubicDimension16LocalWordCarrier`, `HypercubicDimension16ShuffleQuotient`, `HypercubicDimension16ShuffleLinearQuotient`,

`HypercubicDimension16AntisymmetryQuotient`, `HypercubicDimension16ContextualQuotient`, `HypercubicDimension16OrbitCarrierBridge`, `WeightedOrbitBurnside`, `HypercubicDimension16OrbitCharacter`, `HypercubicDimension16LocalLetterCharacter`, `HypercubicDimension16CycleSeedCharacter`, `HypercubicDimension16CycleFamilyCharacter`, `HypercubicDimension16FixedCycleFamilyCarrier`, `HypercubicDimension16CycleBlockDecoration`, `HypercubicDimension16PermutationDecorationTransport`, `HypercubicDimension16PermutationFixedS` `HypercubicDimension16PermutationConjugacy`, `HypercubicDimension16PartitionRepresentatives`, `HypercubicDimension16ConjugacyCompression`).

- **The complete raw $H(4)$ character multiplicities are certified.** The standard signed-axis representation, all 384 trace and determinant values, Reynolds projector, and rank-0–16 character averages are checked. At rank 16 the scalar and pseudoscalar tensor multiplicities are 22,379,179 and 22,368,256; multiplication by the exact active-layout count gives the raw invariant-space dimensions displayed above (`HypercubicRawFDCharacterCensus`, `HypercubicRawFDCharacterCertificate`).

- **The Cartan octic chart has exact conditioning.** The 1287 derivative-free monomials split into 17 surviving signed orbits and negative-stabilizer orbits. The orbit indicators, representative duals, reconstruction theorem, and 17×17 identity matrix are certified (`HypercubicDimension16CartanCensus`, `HypercubicDimension16CartanCertificate`, `HypercubicDimension16CartanBasis`).

- **The Wilson coefficient and classical restoration rate are derived on that chart.** Only the pure-octic coordinate is populated, with coefficient $-a^{12}/20160$. The actual Cartan link observable converges to its quadratic $O(4)$-invariant density with the explicit uniform bound $(5/8)a^4M^4$ (`HypercubicWilsonSymanzikRestoration`).

- **Boundary recombination retains factor one.** The 17 coordinates define a rank-17 degree-zero boundary-cochain quotient; shared-boundary currents cancel exactly. Under the three named analytic bounds, the resulting budget is 5544/5 and the least closing depth is 15 (`HypercubicDimension16BoundarySymanzik`).

- **The frontier is explicit.** The raw character theorem imposes none of the physical relation rows. The derivative-label alpha quotient removes one purely syntactic redundancy. Field-label alpha, adjacent different-owner derivative shuffle, and antisymmetry rows are now equivariant. Innermost Bianchi and on-shell EOM rows are also equivariant, and globally outermost integrated IBP plus traceless one-cycle rows are equivariant. The cross-sector covariant-commutator, traceless anticommutator, and fundamental multitrace rows are now also equivariant. The cycle-index census supplies a literal-group evaluation of its formal expression. Its local-word data, commuting actions, orbit quotient, and sign descent now have a concrete finite realization. The global-to-local derivative map is now an equivariant surjection, and a direct head-matching theorem identifies the full semantic kernel with the equivalence closure of the literal adjacent different-owner generators. On free rational modules, those same literal rows span exactly the kernel of localization. The single-letter field-antisymmetry quotient is now exact and signed $H(4)$-equivariant. Its product lift over every finite family of field slots is also exact: the literal slotwise rows span the full kernel. Its equivariant assembly with the trace context and derivative-shuffle quotient is now exact on the dependent dimension-16 carrier, where the pre-existing position-indexed rows span the full combined kernel. The resulting fixed-sector target is now explicitly equivalent to the finite labeled local-word carrier, compatibly with field relabeling, the signed $H(4)$ action, and tensor sign;

this equivalence descends to the genuine field-label orbit quotients. The signed fixed-point character of that quotient is now defined directly, and a weighted Burnside theorem identifies its factorial multiple with the labeled twisted fixed-point sum. The local cycle character and its exact derivative-degree convolution are now realized by a dependent cycle-seed carrier and proved equal to the recursive partition character. On one positive cycle, the literal family fixed by the relabeling/h equation is now explicitly equivalent to the powered fixed seed with exact preservation of the full tensor-sign product. A dependent positive-cycle-list carrier now performs the full assembly with exact residual derivative degree, and its signed sum is exactly the recursive partition character. The carrier is now realized literally as decorations on a recursive disjoint union of finite blocks under blockwise rotation; the explicit equivalence preserves the full signed weight, so its fixed-decoration character is the same partition character. Canonical block representatives now realize every standard integer partition through eight fields, each actual relabeling permutation is conjugate to its selected representative, and the exact partition-fiber equivalence reduces the Burnside sum to ordinary conjugacy classes. Class cardinality cancels centralizer cardinality and the field factorial, proving equality of the actual orbit character and the cycle-index expression in every sector. The standard rows are now assembled into a single policy-indexed linear operator, before and after the exact contextual quotient. Both actions are genuine rational $H(4)$ representations, the operator is equivariant, and its range is invariant. The normalized operator now descends exactly and equivariantly to the genuine field-label orbit carrier: the literal simultaneous-renaming rows are its full kernel and its range is again invariant. The raw rows are now proved source/successor supported, and exact field filters commute through both normalizations, so the descended operator is block upper-bidiagonal by field count. On the genuine eight-field, zero-derivative orbit module, the signed Reynolds projector and literal character bridge now prove that the scalar invariant target has dimension 11,654. Ordinary two-by-two matrix-index contraction is now proved invariant under simultaneous field relabeling and therefore descends to the genuine orbit carrier. On the implemented pre-quotient sparse module, the singleton traceless rows, adjacent anticommutator rows, and all six rewires of every fundamental trace row are proved to lie in the kernel of this ordinary matrix semantics. The anticommutator proof uses polarized Cayley–Hamilton entrywise; the fundamental proof is the vanishing alternating 3×3 minor with only two matrix-index columns. The checked 98×98 evaluation inverse supplies the matching lower bound, so the complete derivative-free standard trace submodule has finrank 11,556 and its quotient has finrank 98, with explicit exact dual conditioning. This does not close the joint physical quotient: a certified seven-field commutator column has a nonzero eight-field successor component and nonzero Reynolds image, so the field sectors cannot be quotiented independently. Our independent Cartan–Maxwell Hilbert-series socket is now formalized as a bounded fourteen-class character evaluator: its periodic $H(4)$ power traces agree with the literal signed-group data through degree 16, its single-particle resolutions distinguish Bianchi-only and on-shell Maxwell policies, and its cached integer recurrence carries an exact-divisibility ledger for Newton, plethystic, Weyl, and hypercubic averaging. No coefficient row or physical-quotient dimension is claimed from this evaluator here; a bounded certificate replay and a proof that Cartan specialization descends through the full joint relation quotient are still absent. The coordinate-free joint physical relation submodule is now defined before any row ordering, and its off-shell-to-on-shell inclusion is exact. The 11,556 transferred trace rows, one seven-to-eight commutator row, and one four-to-five-field commutator row give a certified joint relation-rank lower bound of 11,558 for either policy; the last is separated by an exact field-five Reynolds coordinate $-1/24$. The remaining finite problem is the complete joint rank and conditioning after Bianchi, EOM, IBP, commutator, and trace rows are assembled across all sectors. The quantitative restoration theorem is restricted to bounded constant Cartan curvature and is not a uniform interacting Symanzik theorem. Constructive Yang–Mills additionally requires a continuum measure, OS reconstruction, and a positive spectral gap.

The remaining inputs

With the checked finite structural components in hand, the route now rests on the following inputs — listed with their exact status in v14, since these are three different situations.

1. **Derive the extraction/recombination constant on the actual map.** *Status: not derived.* Appendix F.5's displayed factors equal 33152 at $b = 2$; Appendix O's different factors with $C_{\text{extract}} = 2$ equal 11088/5. The fluctuation big-O constant, weighted cumulant-generation estimate, rooted tree-graph factor, and basis/dual-jet conditioning are not proved on the actual map. Regrouping connected families by union has norm one; the factor 14 is the local rooted branching count, not a total-contact bound. The third round sharpened what a derivation must overcome: representative-wise truncation is refuted intrinsically, exact norm-below-two extractors exist on the corrected charts, and the transfer of any such extractor to the canonical (F, D) space is refuted by the derivative-blindness and integration-by-parts/block obstructions. Our derivative-aware boundary complex and unpadded raw carrier now repair those two representational defects. The complete raw character census determines the invariant-space dimensions; on the genuine eight-field orbit module the corresponding scalar invariant dimension is now exactly 11,654, and the Cartan octic restriction supplies exact duals and factor-one conditioning. The complete derivative-free eight-field standard trace quotient is separately proved to have dimension 98, with exact evaluation duals and their sharp conditioning constant. Equivariant sparse maps for antisymmetry, Bianchi, on-shell EOM, integrated-by-parts, the covariant-derivative commutator, tracelessness, and the fundamental 2×2 trace identity are now constructed on that carrier. Ordinary two-by-two matrix semantics now annihilates the implemented traceless, adjacent-anticommutator, and fundamental trace rows exactly, and the inverse evaluation certificate proves completeness of their derivative-free quotient. A nonzero seven-to-eight commutator block proves that this result cannot be promoted by treating field sectors independently. *What settles it:* certify the joint cross-sector rank and a dual basis after all standard rows are assembled, without replacing the relations by a desired answer; and extend the classical constant-Cartan $O(a^4)$ estimate to uniform interacting Wilson observables. Then prove the weighted Wilson cumulant/tree estimate and Wilson-RG intertwining law on that same object. The resulting actual constant can then be checked against $C\, b^{3-d_{\max}} \leq 1/3$.

2. **Which support recurrence the glue satisfies.** *Status: v14 asserts both branches of a now-proven dichotomy in different places. What settles it:* one paragraph identifying the recurrence. If coarsen-then-thicken, the uniform bound is already a theorem; if thicken-only, the $O(J)$ growth breaks uniform tree decay and the step needs a new argument.

3. **The two-marked correlation identity.** *Status: missing.* The glue / back-propagation step requires a *two*-marked transport identity of the shape $\langle A; \tau B\rangle_{\mu_0} = \langle QA; \tau' QB\rangle_{\mu_J}$; v14 states the one-marked pushforward duality, and the two-marked version does not follow from it (conditional expectations do not factor). *What settles it:* state it precisely and prove it, or carry it as an explicit named assumption; once stated, its formalization is immediate.

4. **KP convergence at the stopping scale and Wilson-measure reflection positivity.** *Status: abstract results are classical; the route-specific instantiation is open.* One must construct the actual stopping-scale Wilson activities in the same norm, prove the KP criterion uniformly, and verify reflection and boundary-limit compatibility for the same Wilson family and RG kernels.

5. **The real RG map satisfies the recursion.** *Status: the central claimed content of v14's RG analysis; not independently verified here.* Everything above shows the surrounding machine is consistent and sharp; whether the actual Yang–Mills RG map, with its actual constants, satisfies the two-source recursion inside the 1/3-threshold is the mathematical core of the route — the one input that is neither bookkeeping, nor a clarification, nor classical.

Items (1), (2), (3), and (5) contain route-specific mathematics. The classical theorem statements in item (4) do not discharge their Wilson instantiation. The actual RG recursion and constructive-QFT promotion remain open, and the Yang–Mills mass gap remains open.

Lean modules. The results reported here are in `Mettapedia.QuantumTheory.YangMills.RGBootstrap` (the contraction predicate and knife-edge arithmetic), `Mettapedia.QuantumTheory.YangMills.RGCrux` (the even lower-extraction route obstruction), `Mettapedia.QuantumTheory.YangMills.FiniteOSReconstruction`, `Mettapedia.QuantumTheory.YangMills.Z2StrongCouplingGap`, and `Mettapedia.QuantumTheory.YangMills.Z3StrongCouplingGap` (the finite OS reconstruction and strong-coupling gaps), and `Mettapedia.QuantumTheory.YangMills.ProofState` (the continuum-open gating); and, from the second round, `RGBootstrapSlack` (two-source bootstrap and depth trichotomy), `ExtractionMajorant` (displayed-sum audit and generic majorant model), `WilsonBlockMajorant` (the concrete finite Wilson-polynomial realization), `ExtendedExtractionTransfer` (the range-only obstruction and explicit basis/dual-jet transfer theorem), `Dimension16WilsonExtractionCertificate` (the F.4/O.9 relation-policy mismatch, checked source transfer table, and obstructed as-written coordinate certificate), `WilsonPolymerRecombination` (norm-one union regrouping, the total-contact counterexample, and the rooted interpretation of 14), `WilsonExtractionRecombinationConstant` (the two ledger calculations, analytic-factor obstructions, and conditional depth arithmetic), `SupportGrowth` (the recurrence dichotomy), `TransferGapClustering` and `Z2TransferClustering` (gap $\Leftrightarrow$ clustering and the discharged finite continuum gate), `ToyRGMap` (the dynamical realization and depth-15 escape), and `Q8StrongCouplingGap / Q8TransferClustering` (the non-abelian instance); and, from the third round, `SU2WilsonBlockPilot` (the exact SU(2) block, tree gauge, co-tree coordinates, and fixed-boundary Haar reference integral), `SU2CayleyHamiltonTruncationObstruction` (the intrinsic non-descent witness), `SU2MajorantRepairDecision` (the sealed truncation-repair decision), `SU2AssociatedGradedExtraction / SU2AssociatedGradedExtractionDecision` (the graded jet basis, its exact extraction constant, and the dimension separation from the schematic count), `SU2IndependentTraceBasis` (the one-plaquette normal form and its extraction constant), and `V14FDQuotientTransferNoGo` (the derivative-blindness and integration-by-parts/block obstructions and the uninhabited transfer sockets); and, from the fourth round, `V14BoundaryCochainComplex`, `V14BoundaryCochainExtraction`, and `V14BoundaryCochainBootstrap` (the derivative-aware boundary-cochain scaffold and its conditional arithmetic), together with `SU2LatticeFDCensusNoGo` (the exact hypercubic quartic witness and the off-shell/on-shell Lorentz-census no-go); and, from the fifth round, `V14HypercubicFDCensus` (the signed $H(4)$ action and finite raw surface), `V14HypercubicQuarticCensusCertificate` and `V14HypercubicQuarticBasis` (the complete unpadded Cartan quartic census, duals, and identity conditioning), `V14HypercubicQuarticWilsonBridge` (the exact Wilson coordinate vector and nonzero $O(4)$-artifact quotient class), and `V14HypercubicFDCensusPaddingNoGo` (the uninhabited faithful padded dimension-16 exact census); and, from our unpadded dimension-16 construction, `HypercubicUnpaddedFDSyntax` (active dependent syntax), `HypercubicDimension16CartanCensus`, `HypercubicDimension16CartanCertificate`, and `HypercubicDimension16CartanBasis` (the 1287-monomial Cartan octic census, 17 orbit coordinates, duals, and identity conditioning), `HypercubicWilsonSymanzikRestoration` and `HypercubicDimension16BoundarySymanzik` (the actual Wilson coordinate, classical $O(a^4)$ restoration bound, boundary lift, factor-one extraction, and conditional depth arithmetic), and `HypercubicRawFDCharacterCensus`, `HypercubicRawFDCharacterCertificate`, and `HypercubicRawFDDimension16Census` (the complete raw $H(4)$ character certificate and exact active-carrier counts), together with `HypercubicDimension16CompressionCensus`, `HypercubicDimension16CompressionReplayCertificate`, `HypercubicDimension16CompressionSectorCertif` and `HypercubicDimension16CompressionSemanticBridge` (the fourteen-class cycle-index target, literal-group replay, exact Reynolds values, and class-average semantic bridge), together with `HypercubicDimension16ContextualQuotient` and `HypercubicDimension16OrbitCarrierBridge` (the exact contextual relation quotient and its equivariant identification with the finite field-label orbit carrier), `HypercubicDimension16PhysicalRelationOperator` (the off-shell/on-shell joint standard-relation operator, its rational $H(4)$ intertwining law, and its invariant range before and after contextual normalization), and `HypercubicDimension16PhysicalOrbitOperator` (the ordinary field-label quotient map, exact identification of its kernel with the explicit renaming rows, its signed $H(4)$ representation and intertwining law, and the descended physical-relation operator with invariant range), and `HypercubicDimension16PhysicalRelationBlocks` (the exact source/successor field-sector support and its descent through both quotient maps), and `HypercubicDimension16CovariantCompressionWall` (the explicit nonzero seven-to-eight commutator block and the resulting field-count block-diagonal obstruction), together with `HypercubicDimension16JointPhysicalQuotient` (the canonical joint invariant relation preimage, coordinate-free scalar quotient, and exact off-shell/on-shell inclusion), `HypercubicDimension16FDIBPCochainJointBridge` (OUR equivariant semantic bridge from finite derivative-alpha-reduced F, D/IBP cochains onto the joint relation submodule, its policy compatibility, and its exact range theorem), `HypercubicDimension16FieldEightTracePhysicalLift` and `HypercubicDimension16JointTraceRankTransfer` (the actual field-eight lift and its 11,556 independent invariant relation rows), `HypercubicDimension16JointTraceCommutatorRank` (the independent seven-to-eight commutator extension), and `HypercubicDimension16FieldFiveCommutatorSeparator` (the explicit four-to-five-field separator and the joint relation-rank lower bound 11,558), together with `HypercubicDimension16TraceEvaluation` (ordinary matrix-index contraction, field-relabel descent, and exact kernel theorems for the three implemented trace-relation families), `HypercubicDimension16SignedOrbitCoinvariants`, `HypercubicDimension16TraceOrbitBasis`, and `HypercubicDimension16TraceRankTheorem` (the independently defined signed quotient, its checked orbit basis, and exact sparse row rank), `HypercubicDimension16PhysicalTraceWitnessBridge`, `HypercubicDimension16PhysicalTraceEvaluationCertificate`, and `HypercubicDimension16PhysicalTraceEvaluationReynolds` (literal-row semantics, the exact 98×98 inverse, and genuine Reynolds normalization), and `HypercubicDimension16PhysicalTraceQuotient`, `HypercubicDimension16PhysicalTracePublicConditioning`, and `HypercubicDimension16TraceEvaluationBootstrap` (the complete derivative-free standard trace quotient, exact public duals, sharp conditioning, and conditional depth arithmetic), and `WeightedOrbitBurnside` with `HypercubicDimension16OrbitCharacter` (the general weighted quotient identity and its specialization to the actual signed orbit character), together with `HypercubicDimension16LocalLetterCharacter` and `HypercubicDimension16CycleSeedCharacter` (the exact local cycle factor and its dependent derivative-degree convolution), and `HypercubicDimension16CycleFamilyCharacter` (the one-cycle fixed-family equivalence and exact tensor-sign product), together with `HypercubicDimension16FixedCycleFamilyCarrier` (the dependent positive-cycle-list assembly and exact residual-degree character), and `HypercubicDimension16CycleBlockDecoration` (the concrete disjoint-cycle permutation, fixed-decoration equivalence, and signed character identity), together with `HypercubicDimension16PermutationDecorationTransport`, `HypercubicDimension16PermutationFixedSector`, `HypercubicDimension16PermutationConjugacy`, `HypercubicDimension16PartitionRepresentatives`, and `HypercubicDimension16ConjugacyCompression` (transport to arbitrary field permutations, exact partition representatives, conjugacy-fiber regrouping, factorial cancellation, and equality with the actual orbit character) — all in the same namespace, all axiom-guarded to {`propext`, `Classical.choice`, `Quot.sound`} with no `sorry`. The manuscript analyzed here is Ben Goertzel's *Yang–Mills Mass Gap in 4D: RG Bootstrap, Polymer Expansion, OS Reconstruction* (v14); constructions explicitly labeled ours are independent formal repairs within the same proof programme, not claims about content already present in that manuscript.